



Redox-active organic polymers: emerging materials for sustainable energy storage and harvesting

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Abstract

Redox-active organic polymers (RAOPs) have emerged as a versatile and sustainable class of materials for next-generation energy technologies. Their unique ability to undergo reversible redox reactions, combined with lightweight, flexible, and tunable structures, makes them highly suitable for diverse applications. This review presents a comprehensive overview of recent advancements in RAOPs for energy storage, energy harvesting, and catalysis. In storage systems, RAOPs offer promising alternatives to inorganic materials in batteries and supercapacitors, including all-organic configurations. Their integration into organic photovoltaics and thermoelectric devices highlights their role in efficient energy conversion and hybrid energy systems. In catalysis, RAOPs enable enhanced performance in hydrogen evolution, CO₂ reduction, and photocatalytic processes. We discuss molecular design strategies, structure–property relationships, and synthetic approaches that govern RAOP functionality, while also addressing key challenges such as stability, conductivity, and scalability. With growing interest in eco-friendly, high-performance materials, RAOPs hold transformative potential to advance sustainable and self-powered energy systems.

Keywords Redox-active organic polymers (RAOPs) · Sustainable energy storage · Organic photovoltaics · All-organic batteries

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Abbreviations

AOBs	All-Organic Batteries
AFM	Atomic Force Microscopy
APBs	Aqueous Proton Batteries
AQ	Anthraquinone
BDT-T	Benzodithiophene-Thiophene
BHJ	Bulk Heterojunction
BPY	4,4'-Bipyridine
Bro-PCN	Brown Polymeric Carbon Nitride
BT	Benzothiadiazole
CBR	Carbazole-Based Non-Fullerene Acceptor
CMPs	Conjugated Microporous Polymers
CN-TH	Cyano/Cyanamide-Modified Carbon Nitride
COF	Covalent Organic Framework
CoTAPc	Cobalt-Tetraaminophthalocyanine
CMPs	Conjugated Microporous Polymers
CONASHs	Bis(Diimino)Nickel Coordination Nanosheets
COPs	Covalent Organic Polymers
CNTs	Carbon Nanotubes
DAC	Direct Air Capture
D-A	Donor-Acceptor
DARp	Direct C–H Arylation Polymerization
DBTh	Benzodithiophene-4,8-Dione
DFT	Density Functional Theory
DPZ	2,3-Diaminophenazine
DOC	Direct Ocean Capture
EDCC	Electrochemical Direct CO ₂ Capture
EDOT	3,4-Ethylenedioxythiophene
EET	Extracellular Electron Transfer
EIS	Electrochemical Impedance Spectroscopy
F4TCNQ	2,3,5,6-Tetrafluoro-7,7,8,8-Tetracyanoquinodimethane
FM	A Polymer Acceptor Material
FRET	Fluorescence Resonance Energy Transfer
FSBM	Fluorene-Silafluorene-Based Non-Fullerene Acceptor
FT-IR	Fourier Transform Infrared Spectroscopy
HCPCs	Hydrogen-Bonding Conductive Polymer Catalysts
HER	Hydrogen Evolution Reaction
HOMO	Highest Occupied Molecular Orbital
IPOP	Ionic Porous Organic Polymer
LIBs	Lithium-Ion Batteries
MOF	Metal-Organic Framework
Mo ₂ C	Molybdenum Carbide
MoSe ₂	Molybdenum Selenide
MOP	Metal-Organic Polymer
NF	Nickel Foam
NFAs	Non-Fullerene Acceptors

NiTAPc	Nickel-Tetraaminophthalocyanine
NHP	Azo-Coupled Hyperbranched Polymers
NTCDA	Naphthalene-1,4,5,8-Tetracarboxylic Acid
OPECs	Organic Photoelectrochemical Cells
OPVs	Organic Photovoltaics
OSCs	Organic Solar Cells
PA	Poly-Adenine
PANI	Polyaniline
PCE	Power Conversion Efficiency
PCBM	Phenyl-C61-Butyric Acid Methyl Ester
PCN	Polymeric Carbon Nitride
PCNDs	Polymeric Carbon Nitride Dots
PCNNs	Polymeric Carbon Nitride Nanosheets
PDQPZ	Poly(Diquinoxalino-Phenazine)
PDA	Polydopamine
PEDOT	PSS: Poly(3,4-Ethylenedioxythiophene)-Poly(Styrenesulfonate)
PET	Polyethylene Terephthalate
PNMZ	Ladder-Type Redox-Active Polymer
POP	Porous Organic Polymers
PQ	Phenanthrenequinone
PVA	Polyvinyl Alcohol
PVDF	Polyvinylidene Difluoride
PVPP	Polyvinylpolypyrrolidone
QE	Ester-Functionalized Quinoxaline
RAOPs	Redox-Active Organic Polymers
RGO	Reduced Graphene Oxide
ROSs	Reticular Organic Skeletons
SFBM	Silafluorene-Based Non-Fullerene Acceptor
SIBs	Sodium-Ion Batteries
TEG	Thermoelectric Generator
TSN	Tunicate Sulfated Nanocellulose
UV-Vis	Ultraviolet-Visible Spectroscopy
XPS	X-ray Photoelectron Spectroscopy
ZOBs	Zinc-Organic Batteries
zT	Thermoelectric Figure of Merit

Introduction

In the pursuit of sustainable energy solutions, the demand for lightweight, cost-effective, and environmentally friendly materials has intensified across multiple sectors. Among the various emerging materials, redox-active organic polymers (RAOPs) have garnered significant attention due to their unique combination of redox functionality, tunable structures, and compatibility with green chemistry principles [1]. Unlike conventional inorganic materials, which often rely on rare or toxic elements, RAOPs are composed primarily of earth-abundant elements such as carbon, hydrogen, nitrogen,

and oxygen, making them highly appealing for the development of next-generation energy technologies [2]. The concept of redox-active polymers traces back to early developments in conducting polymers in the 1970s, such as polyacetylene, polyaniline, and polypyrrole, which demonstrated electron transport via conjugated systems [3]. These early studies laid the foundation for the development of RAOPs, which combine redox functionality with structural versatility, processability, and sustainability. While the field has advanced significantly over the past decades, a comprehensive understanding of how molecular design impacts performance across different energy domains remains fragmented.

This review aims to address this gap by presenting a focused and critical overview of recent advances in RAOPs, specifically in the domains of energy storage, energy harvesting, and catalysis. RAOPs are now revolutionizing these fields due to their unique electrochemical properties and versatility [4–6]. These polymers can undergo reversible oxidation and reduction (redox) reactions, making them indispensable in various technological applications [7]. Their ability to donate or accept electrons in a controlled manner is attributed to conjugated systems, functional groups, and heteroatoms that facilitate electron transfer [8, 9]. This property allows RAOPs to be tailored for specific needs, enhancing their potential for integration into next-generation technologies [10]. Their light weight, tunable structure, and inherent sustainability make RAOPs promising candidates for use in energy storage devices like batteries and supercapacitors, as well as in energy-harvesting systems and catalytic processes [11–13].

Recent advancements in polymer chemistry have enabled precise control over molecular weight, morphology, and functionalization through techniques such as chemical polymerization, electrochemical polymerization, and template-assisted synthesis [14, 15]. These synthetic strategies have opened new possibilities for creating RAOPs with tailored redox properties, essential for improving the performance of energy devices, catalytic systems, and other applications [16, 17].

RAOPs have emerged as sustainable and versatile components in advanced energy storage systems. Their organic frameworks, infused with redox-active functional groups such as quinones, nitroxides, or conjugated systems, facilitate reversible electron transfer during charge and discharge cycles [18]. This makes them highly effective in devices ranging from batteries to supercapacitors. RAOPs provide flexibility and processability, allowing for the creation of lightweight, flexible, and even stretchable energy storage devices [19]. This is especially advantageous for use in portable electronics, wearable technology, and electric vehicles, where the need for lightweight and flexible power sources is crucial [20].

Energy harvesting is another area where RAOPs show tremendous promise [21]. Energy harvesting involves converting ambient energy from the environment, such as solar, thermal, or mechanical energy, into usable electrical energy [22–25]. The tunable electronic properties of RAOPs allow them to be used in devices like organic photovoltaics (OPVs), thermoelectric generators, and piezoelectric materials [26]. In OPVs, RAOPs can serve as the active layer that absorbs sunlight and generates charge carriers, while in thermoelectric devices, they can efficiently convert heat into electricity [27]. The convergence of energy harvesting and storage within RAOP-

based platforms offers an exciting pathway toward the development of self-powered, miniaturized, and eco-friendly energy systems.

Beyond energy storage and harvesting, RAOPs are gaining traction in catalysis, particularly in electrochemical and photocatalytic reactions [28]. Catalysis plays a critical role in industrial processes, environmental protection, and energy conversion [29]. RAOPs can act as efficient catalysts or catalyst supports due to their redox-active sites, which enable electron transfer and enhance reaction rates [30]. Furthermore, their porous structure and high surface area make them ideal for immobilizing catalytic species, thereby improving their stability and reusability [31, 32]. This property is crucial for designing sustainable catalytic systems with tailored characteristics for various chemical processes.

In electrochemical energy storage, RAOPs have led to breakthroughs in the performance of batteries and supercapacitors. For example, in lithium-ion batteries, RAOPs can be incorporated into the cathode, where their redox activity facilitates the reversible insertion and extraction of lithium ions, resulting in high energy density and long cycle life [33]. Similarly, in supercapacitors, RAOPs have been utilized as electrode materials to enhance capacitance and energy storage capabilities [34]. The fast and reversible redox reactions in RAOPs make them ideal for high-power applications that require rapid charge and discharge cycles [35]. These properties make RAOP-based energy storage devices highly competitive for a variety of industrial and consumer applications.

All-organic batteries (AOBs), which rely solely on organic materials for both the anode and cathode, represent one of the most promising uses of RAOPs in energy storage [36]. The development of AOBs addresses several challenges associated with conventional inorganic batteries, including resource sustainability, cost, and safety [37]. RAOPs, with their unique combination of high capacity and low cost, offer a more sustainable alternative for energy storage [38–40]. The flexibility of organic materials also allows for the design of batteries that can be molded into various shapes and sizes, further enhancing their potential for integration into flexible and portable electronic devices [41].

The integration of RAOPs into energy-harvesting devices like solar cells and thermoelectric generators offers an exciting opportunity for the creation of multifunctional systems that both generate and store energy [42]. In organic photovoltaics, RAOPs can be engineered to absorb a broad range of solar radiation, converting it into electrical energy [43, 44]. In thermoelectric devices, RAOPs can efficiently harvest waste heat, contributing to energy recovery in industrial and electronic systems [45]. The combination of energy harvesting and storage in RAOP-based devices could lead to the development of self-powered systems that are capable of continuous operation without external power sources [46].

RAOPs also demonstrate great potential in photocatalysis, a process in which light energy is used to drive chemical reactions [47]. For instance, RAOPs have been explored as photocatalysts for water splitting, CO₂ reduction, and other environmentally significant processes [48, 49]. The unique redox properties of these polymers enable them to participate in photocatalytic reactions that require the transfer of electrons under light irradiation [50]. The ability to harness solar energy for chemical

transformations makes RAOPs a promising solution for sustainable energy production and environmental remediation [51, 52].

The development of redox-active organic catalysts has also led to improvements in catalytic processes, particularly in the fields of organic synthesis and environmental protection [53]. RAOPs can be used as molecular catalysts or as supports for metal nanoparticles, where their redox-active sites enhance catalytic efficiency and selectivity [54]. These properties are especially useful in reactions such as hydrogenation, oxidation, and carbon-carbon coupling, where high selectivity and efficiency are essential [55, 56]. The versatility of RAOPs as both catalysts and catalyst supports offers new possibilities for designing more efficient and sustainable catalytic systems [57].

The potential of RAOPs is not limited to energy storage, harvesting, and catalysis [58]. Their inherent sustainability and the ability to be synthesized from abundant and renewable resources make them ideal candidates for addressing global challenges related to resource depletion and environmental impact [59]. The integration of RAOPs into various energy systems, coupled with their ability to be tailored for specific applications, positions them as critical materials in the development of next-generation energy technologies.

This review aims to critically evaluate the recent progress in the field of redox-active organic polymers, focusing on their applications in energy storage (batteries, supercapacitors), energy harvesting (organic photovoltaics, thermoelectrics), and catalytic processes (electrocatalysis and photocatalysis). We highlight how advances in molecular design, synthetic strategies, and hybrid material systems have contributed to overcoming the intrinsic limitations of RAOPs, such as low conductivity and stability. In doing so, we offer an integrated perspective on how RAOPs are shaping the landscape of sustainable energy technologies and outline future directions for research in this transformative area.

Advancements of redox-active organic polymers in energy and catalysis

RAOPs represent a revolutionary class of materials that have rapidly advanced in the fields of energy storage, energy harvesting, and catalysis [60]. These materials are distinguished by their unique electrochemical properties, which allow them to undergo reversible redox reactions, making them highly versatile for various applications [61–64]. Their tunable molecular structures and sustainability give RAOPs a significant edge over traditional inorganic materials, positioning them as promising candidates for the future of energy technologies [65]. The history of RAOPs dates back to the mid-20th century when the first conductive organic polymers, such as polyacetylene, were discovered [66]. This laid the groundwork for further developments, eventually leading to the emergence of redox-active polymers [67]. Over the years, breakthroughs such as the discovery of polyaniline and its redox capabilities have demonstrated the potential for organic materials to replace inorganic components in energy devices and catalysis realm of energy storage, RAOPs have shown immense potential, particularly as alternatives to conventional lithium-ion batteries

[68, 69]. Unlike inorganic materials such as lithium cobalt oxide, RAOPs, including polythiophene and polyaniline, are lightweight, flexible, and capable of efficient charge storage through reversible redox processes [70].

Organic radical batteries (ORBs), which utilize RAOPs as cathode materials, have demonstrated high energy densities and fast charge-discharge rates [71]. Their use in flexible and wearable electronics is particularly promising, as RAOPs can be processed into films or thin layers suitable for integration into flexible devices [72–75]. These innovations are helping drive the development of next-generation energy storage systems that are not only more efficient but also more sustainable [76]. They have made significant contributions to energy harvesting, particularly in organic photovoltaics (OPVs) and thermoelectric devices [77]. In OPVs, RAOPs act as both light-absorbing and charge-transporting materials, converting sunlight into electricity. The development of donor-acceptor polymers and other molecular design strategies has greatly enhanced the power conversion efficiency of OPVs [78–80]. Similarly, RAOPs have been employed in thermoelectric devices, where their ability to efficiently convert waste heat into electricity offers a sustainable solution to energy harvesting [81–84]. The flexibility and tunability of RAOPs make them ideal candidates for integrating into a wide range of renewable energy technologies [85]. In addition to their role in energy storage and harvesting, RAOPs are playing an increasingly important role in catalysis [86]. Redox-active polymers are now being explored as catalysts or catalyst supports for various reactions, including hydrogen evolution, oxygen reduction, and carbon dioxide reduction [87]. Functionalized RAOPs, particularly those with transition metal complexes, have been shown to mimic the catalytic sites of natural enzymes, enabling efficient and selective catalytic processes [88]. These advances have significant implications for fuel cells, water splitting, and environmental remediation, where both high performance and sustainability are crucial [89–91].

The synthesis and characterization of RAOPs have progressed considerably in recent years [92]. Advanced techniques like controlled radical polymerization and post-polymerization modifications enable precise control over the structure and functionality of these materials [93]. This has facilitated the development of RAOPs with optimized redox potentials, improved conductivity, and enhanced stability [94]. Advanced analytical tools, such as in situ spectroscopy and computational modeling, are offering deeper insights into the redox mechanisms of RAOPs. These insights aid researchers in designing materials with improved performance for specific applications [95, 96]. A key area of research in RAOPs is the integration of these materials with other substances to create hybrid systems [97]. Combining RAOPs with inorganic nanoparticles or carbon-based materials has led to composite materials with superior electrochemical performance and mechanical properties [98]. For example, hybrid materials have been used to create high-capacity electrodes with improved cycling stability, and in catalysis, the combination of RAOPs with conductive supports has led to highly efficient electrocatalysts [99]. These hybrid systems showcase the potential of RAOPs to complement and enhance the properties of other materials, propelling the advancement of cutting-edge energy storage devices and catalytic systems [100–103].

Despite their many merits, the widespread adoption of RAOPs faces several challenges [104]. Long-term stability, scalability, and cost-effectiveness remain significant barriers.

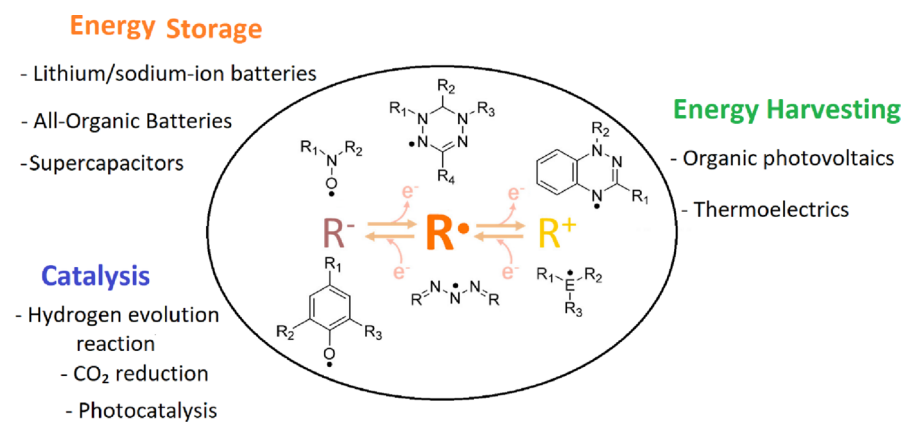
ers to their full commercialization. For instance, while RAOPs demonstrate excellent electrochemical properties, their stability during prolonged cycling or under harsh conditions can be a limiting factor [105]. Researchers are exploring various strategies to improve the durability of these materials, such as cross-linking polymer chains and incorporating stabilizing additives [106]. Additionally, scaling up the synthesis of RAOPs and ensuring their environmental sustainability are critical areas of focus [107]. One of aspects of RAOPs is their environmental sustainability. As the world shifts toward a circular economy, the use of renewable resources to produce RAOPs offers a sustainable alternative to traditional petroleum-based materials [108]. RAOPs can be designed to be biodegradable or recyclable, which helps reduce their environmental impact. Moreover, their role in energy storage and harvesting can contribute to the reduction of greenhouse gas emissions by enabling the more efficient use of renewable energy sources [109].

Looking ahead, the potentials of RAOPs are vast. Emerging fields such as bioelectronics and smart materials, which respond to external stimuli, stand to benefit greatly from the unique properties of these polymers [110]. The integration of artificial intelligence and machine learning in material discovery will likely accelerate the development of novel RAOPs with tailored properties. As interdisciplinary research continues to advance, RAOPs are expected to play an increasingly important role in addressing global challenges related to energy, catalysis, and sustainability [111–114]. Redox-activolymers have come a long way since their inception, evolving from simple laboratory curiosities to practical materials with transformative applications in energy storage, energy harvesting, and catalysis [115]. These materials are revolutionizing the way we think about sustainable technologies and have the potential to replace traditional inorganic components in many critical applications [116]. While challenges remain, the future of RAOPs looks promising, with continued research likely to unlock their full potential and pave the way for the next generation of green technologies [117]. A schematic summary of the key advancements and application domains of redox-active organic polymers is illustrated in Scheme 1 to provide a simplified visual overview of the concepts discussed in this section.

Key applications of redox-active organic polymers in energy storage, energy harvesting, and catalysis

Energy storage

RAOPs have emerged as a focal point in materials science due to their potential as sustainable and versatile components in advanced energy storage systems [118]. Defined by organic frameworks infused with redox-active functional groups, such as quinones, nitroxides, or conjugated systems, RAOPs facilitate reversible electron transfer during charge and discharge cycles, rendering them highly effective in devices ranging from batteries to supercapacitors [119]. Composed of abundant elements like carbon, hydrogen, and oxygen, these polymers present a cost-effective and environmentally friendly alternative to conventional inorganic electrode materials [120]. Their tunable electrochemical properties and structural adaptability further



Advancements and Applications of Redox-Active Organic Polymers

Scheme 1 Simplified schematic illustrating the main application areas of RAOPs: energy storage, energy harvesting, and catalysis

enhance their appeal, positioning RAOPs as key players in the pursuit of next-generation energy storage technologies [121].

One of the most prominent applications of RAOPs lies in rechargeable batteries, including lithium-ion batteries (LIBs) and emerging post-lithium systems such as sodium-ion (SIBs) and potassium-ion batteries. Depending on their redox potential and molecular architecture, RAOPs can function as either cathodes or anodes [122]. For instance, polymers featuring conjugated backbones and pendant redox groups have exhibited high specific capacities and excellent rate capabilities, making them ideal for batteries requiring elevated energy density and rapid charge/discharge cycles. A notable example of this potential is illustrated by Ma et al. [123], who explored the integration of carbon nanotubes (CNTs) with azo-coupled redox-active polymers to create core-shell structured cathodes for LIBs. Addressing the inherent limitations of RAOPs, such as poor intrinsic conductivity and low practical capacity, the researchers encapsulated CNTs within azo-coupled hyperbranched polymers (NHP) via in situ copper(I)-catalyzed oxidative coupling condensation, forming NHP@CNT composites. These composites, with conductive CNT cores encased in electrochemically active NHP shells, demonstrated superior reversible capacity, rate capability, and cycling stability compared to standalone NHP cathodes. The enhanced performance stemmed from improved electron transport, greater exposure of active sites, and better electrolyte accessibility, underscoring the efficacy of core-shell architectures in advancing sustainable, high-performance organic electrodes for LIBs.

Beyond LIBs, RAOPs have shown significant promise in post-lithium technologies, exemplified by the work of Pan et al. [124], who focused on enhancing the energy density of organic cathode materials for SIBs. The study introduced two dihydrophenazine-based organic polymer cathodes, DPZPC and DPZPA, synthesized through Buchwald-Hartwig condensation polymerization. By designing planarized conjugated p-type polymers with multi-redox-active centers, the researchers tack-

led the challenges of low specific capacity and poor electronic conductivity typical of traditional p-type organic materials. The DPZPC polymer, distinguished by its planarized conjugated repeat units, outperformed DPZPA with higher electronic conductivity, increased surface area, and improved ionic transportation. This resulted in exceptional electrochemical performance, including high specific capacity, long cycling stability, and robust rate capability. Such molecular engineering highlights RAOPs' potential as sustainable, high-energy-density cathode materials for next-generation rechargeable batteries, particularly in the context of sodium-based systems.

In a significant contribution to the design of stable radical-based energy materials, Su et al. introduced a class of porous radical polymers engineered via an electronic isolation strategy to maintain persistent open-shell states [125]. By spatially and electronically decoupling the radical moieties from the π -conjugated backbone, the authors achieved materials with exceptional thermal and redox stability, mitigating radical–radical recombination. These PRPs exhibited long-lived radical activity and electrochemical reversibility, which are critical for applications in high-performance organic electrodes, supercapacitors, and even spintronic devices. This approach highlights the role of molecular design in stabilizing redox-active polymers while preserving their electrochemical accessibility and conductivity.

RAOPs also extend their utility to supercapacitors, where rapid charge/discharge cycles and high power density are paramount. Zhan et al. [126] investigated this application by developing redox-active conjugated microporous polymers (CMPs) based on benzodithiophene-4,8-dione (DBTh) for high-performance supercapacitor energy storage. Through a one-pot Suzuki coupling polymerization, they synthesized two novel CMPs, DBTh-TPA and DBTh-TPT, designed to overcome the limitations of traditional CMPs, such as low electronic conductivity and insufficient redox-active sites. The incorporation of DBTh units into the polymer backbone enhanced conductivity, charge transfer efficiency, and faradaic energy storage capabilities. With high thermal stability, significant surface areas, and hierarchical porous architectures, these CMPs, particularly DBTh-TPT, demonstrated remarkable specific capacitances and cycling stability in both three-electrode and symmetric two-electrode supercapacitor setups. This study underscores the value of embedding redox-active moieties into CMP frameworks, offering a viable pathway for advancing supercapacitor technologies and hybrid devices that merge battery and supercapacitor traits to optimize both energy and power densities. The versatility of RAOPs is further enhanced by their lightweight and mechanically flexible properties, which reduce the overall weight of energy storage devices, crucial for portable electronics and electric vehicles, and enable applications in flexible, wearable technologies [127–130]. Their compatibility with aqueous electrolytes also promotes safer, more sustainable systems compared to those reliant on flammable organic electrolytes. However, RAOPs face significant hurdles that impede their broad adoption. Low electronic conductivity, solubility in electrolytes, and diminished stability over repeated cycles remain critical challenges. To address these, ongoing research has explored strategies such as integrating conductive additives, crosslinking polymer chains, and developing nanostructured architectures. The lightweight and flexible nature of RAOPs, combined with their sustainability, positions them as transformative materials, provided these limitations can be surmounted through innovative design. In conclusion, RAOPs represent a

promising frontier in energy storage, offering tunable properties and sustainable compositions that align with the demands of modern technology. Case studies, such as those by Ma et al., Pan et al., and Zhan et al., illustrate the tangible progress in overcoming RAOPs' limitations through sophisticated material engineering—whether via core-shell composites for LIBs, multi-redox-active polymers for SIBs, or porous CMPs for supercapacitors.

Building on the advancements in RAOPs, recent studies have further demonstrated the potential of organic polymers in energy storage applications. For instance, Wang et al. [131] synthesized a novel electron-delocalized organic polymer, PNZI, through a facile one-step polymerization process using 2,3-diaminophenazine (DPZ) and naphthalene-1,4,5,8-tetracarboxylic acid (NTCDA) as precursors. The study aimed to address the limitations of organic materials in aqueous electrochemical energy storage, such as inadequate active sites, low conductivity, and solubility issues (Fig. 1A). The PNZI polymer was designed to integrate redox-active functional groups ($C=O$ and $C=N$) and exhibit enhanced conjugation, ensuring both structural stability and excellent electronic conductivity. The polymer demonstrated remarkable performance as an electrode material for aqueous proton batteries (APBs) without the need for conductive additives, showcasing high reversible specific capacity, outstanding

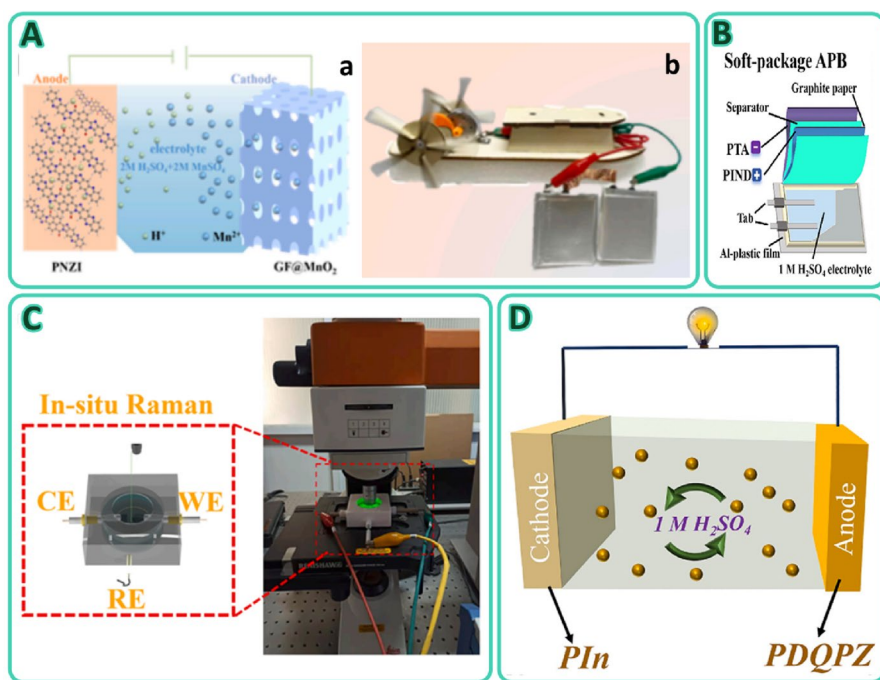


Fig. 1 (A) (a) Schematic diagram of the operating mechanism of PNZI//GF@MnO₂ proton battery. (b) Photograph of the APB. Reprinted with permission from [131]. (B) Schematic diagram of the soft-package APB based on PTA anode and PIND cathode. Reprinted with permission from [132]. (C) Schematic illustration and digital photo of the in-situ Raman measurement. Reprinted with permission from [134]. (D) Sketch map of the electrochemical process in PIn//PDQPZ soft-package APB. Reprinted with permission from [134]

rate capability, and cycling durability over 5000 cycles. The authors employed a combination of structural analysis and in situ/ex situ characterization techniques to elucidate the proton migration pathway and redox mechanism within the PNZI structure. Additionally, they constructed high-performance full cells, including an all-polymer aqueous proton battery and a PNZI//MnO₂ full cell, achieving significant energy density and cycle life. This work provides valuable insights into the design of highly conductive organic polymer electrodes and their application in high-performance aqueous energy storage systems. In a similar vein, Yang et al. [132] developed a dual redox-active organic polymer (PTA) designed for aqueous proton batteries (APBs), aiming to enhance proton storage capabilities and address the limitations of current organic materials in energy storage applications. The study focused on creating a polymer with extended π -conjugated frameworks incorporating dual redox-active moieties, which not only improved redox activity and electronic properties but also ensured structural integrity (Fig. 1B). The PTA polymer demonstrated exceptional proton storage capacity and long-term stability, maintaining a high capacity retention over numerous cycles. In situ techniques and theoretical calculations revealed efficient redox processes at C = N and C = O active sites during proton uptake and release. The researchers also assembled a soft-package APB device using the PTA electrode, which exhibited impressive electrochemical performance and operational lifespan, highlighting its potential for practical energy storage applications. This work contributes to the advancement of sustainable polymer electrodes for cost-effective and high-performance rechargeable aqueous batteries. Expanding the scope of organic polymers, Saber et al. [133] explored the development of heteroatom-rich fluorene-based benzoxazine-linked porous organic polymers (POPs) as potential candidates for energy storage applications. The study focused on synthesizing two novel POPs, FI-TPA and FI-Cz, through a one-pot Mannich coupling reaction, leveraging the unique properties of fluorene derivatives and benzoxazine linkages. The authors aimed to address the growing demand for efficient energy storage systems by designing materials with high porosity, redox activity, and conductivity. Comprehensive characterization techniques, including FTIR, solid-state NMR, and nitrogen sorption analysis, confirmed the structural integrity and porosity of the synthesized polymers. Electrochemical evaluations revealed that the FI-TPA polymer exhibited superior specific capacitance, energy density, and cycle stability compared to FI-Cz, attributed to its higher microporosity and specific surface area. The study underscores the potential of heteroatom-rich POPs as advanced electrode materials for supercapacitors, offering a scalable and cost-effective approach to energy storage solutions. This work contributes to the ongoing research in sustainable energy technologies by providing insights into the design and optimization of porous organic materials for enhanced electrochemical performance. Further advancing the field, He et al. [134] designed and synthesized a highly redox-active polymer, poly(diquinoxalino-phenazine) (PDQPZ), with extended imine-conjugated backbones for use in aqueous proton batteries (APBs) (Fig. 1C). The study aimed to address the limitations of organic polymers in APBs, such as low redox capacity and electron affinity, by optimizing molecular orbitals and enhancing electron delocalization. The PDQPZ polymer demonstrated exceptional proton storage capabilities, including high capacity and cycling stability, supported by theoretical calculations and in-operando moni-

toring techniques. The researchers also fabricated a high-performance APB device using PDQPZ, which exhibited long-term stability across a wide temperature range (Fig. 1D). This work highlights the potential of molecularly engineered polymers for developing efficient and durable energy storage systems.

In another innovative approach, Zhang et al. [135] explored the development of advanced zinc-organic batteries (ZOBs) by designing reticular organic skeletons (ROSs) with multiple protophilic redox-active sites for superior proton storage. The study focused on overcoming the limitations of traditional Zn^{2+} charge carriers, such as sluggish kinetics and structural instability, by utilizing protons (H^+) as an alternative charge carrier due to their small size and rapid redox kinetics. The researchers synthesized ROSs through the polymerization of 2,6-diaminoanthraquinone and 2,4,6-triformylphloroglucinol, creating a nanofibrous structure stabilized by intermolecular hydrogen bonding and π - π stacking interactions. This design enabled the exposure of multiple carbonyl sites, facilitating efficient proton storage and rapid electron delocalization. The resulting $\text{Zn}||\text{ROSs}$ battery demonstrated exceptional electrochemical performance, including high capacity, excellent rate capability, and long-term cycling stability. The study also revealed that the superior performance was driven by a high-kinetics, proton-coupled five-electron redox process. This work highlights the potential of protophilic ROSs in advancing the development of high-performance ZOBs, offering a promising pathway for next-generation energy storage systems. Complementing these efforts, Yan et al. [136] developed a redox-active covalent organic framework (COF) with aniline-fused quinonoid units, termed TPAD-COF, as a cathode material for aqueous proton batteries (Fig. 2A). The study aimed to address the challenges of electrode material corrosion and dissolution in acidic electrolytes, which hinder the practical application of aqueous proton batteries. The TPAD-COF cathode demonstrated high capacity and long-term cycling stability, attributed to its multiple redox-active centers and superhydrophilic properties, which facilitate efficient proton transport. Comprehensive *ex situ* spectroscopy and density functional theory (DFT) calculations revealed that both the $-\text{NH}-$ and $\text{C}=\text{O}$ groups in the aniline-fused quinonoid units exhibit prominent redox activity, enabling efficient proton storage (Fig. 2B). DFT is a quantum mechanical method that calculates the electronic structure of matter by describing its properties as a function of the electron density rather than the many-electron wavefunction [137–140]. The authors also assembled an all-organic proton battery using TPAD-COF as the cathode and anthraquinone (AQ) as the anode, which showed promising performance, including good rate capability and cycling stability. This work highlights the potential of redox-active COFs as durable and high-capacity electrode materials for aqueous proton batteries, offering a new direction for the design of sustainable energy storage systems. Continuing the trend of molecular innovation, He et al. [141] explored the development of a novel ladder-type redox-active polymer, PNMZ, designed for ultra-stable and fast proton storage in aqueous proton batteries (APBs). The study focused on addressing the limitations of organic polymers in proton storage by leveraging a covalent cycloconjugation conformational strategy to create a rigid-coplanar polymer with extended $\pi\pi$ -electron delocalization (Fig. 2C). This structural design facilitated rapid intra-chain charge transport and enhanced proton-storage redox kinetics, as evidenced by *in-operando* monitoring techniques and theoretical calculations. The

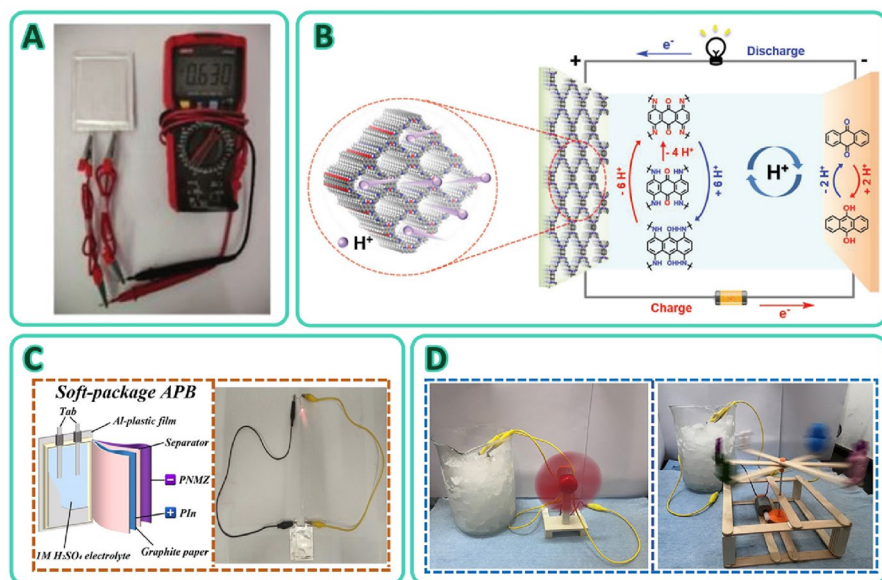


Fig. 2 (A) Optical voltage measurements of the TPAD-COF//AQ all-organic punch battery. Reprinted with permission from [136]. (B) Illustration of the working mechanism for TPAD-COF/AQ all-organic proton battery in a 1.2 M H₂SO₄ electrolyte. Reprinted with permission from [136]. (C) Illustration of device structure. Reprinted with permission from [141]. (D) Digital photos showing the efficient power supply of the APB device at a low-temperature state of -10°C . Reprinted with permission from [141]

PNMZ polymer demonstrated exceptional proton-storage capacity and long-term cycling stability, maintaining high capacity retention over numerous cycles. The researchers successfully constructed an all-polymer APB device, showcasing high energy density and remarkable cycle performance, thereby highlighting the potential of PNMZ for practical applications in efficient and stable energy storage systems (Fig. 2D). The work underscores the importance of molecular design in advancing the performance of proton batteries, contributing to the broader goal of developing sustainable energy storage solutions.

A structurally distinct approach to redox-active frameworks was reported by Su et al., who synthesized benzofuran-linked COFs via irreversible cascade reactions, resulting in highly crystalline, porous, and thermally robust materials [142]. The benzofuran linkages introduced extended π -conjugation and electronic delocalization, enhancing the intrinsic conductivity and redox tunability of the framework. These properties are particularly desirable for electrochemical applications such as charge storage and electrode design. This study exemplifies how the deliberate incorporation of stable conjugated motifs within COFs can yield redox-active architectures with synergistic structural and electronic properties, expanding the scope of RAOPs in next-generation energy materials. Moreover, Wang et al. recently reported a novel class of π -conjugated radical polymers that combine semiconducting behavior with high redox activity for application in ambient aqueous batteries [143]. These materials integrate stable nitroxide radicals with π -conjugated backbones, enabling concurrent ionic and electronic transport in neutral aqueous electrolytes. The optimized

balance between radical density and conjugation facilitates rapid charge–discharge cycles, excellent cycling stability, and efficient electron/hole delocalization. Importantly, the semiconducting nature of the conjugated backbone enhances long-range charge transport, making these polymers promising candidates for safe, high-performance energy storage systems under environmentally benign conditions. This approach underscores the importance of tailoring both the redox-active units and the electronic structure in RAOPs to realize multifunctional energy materials operable in aqueous environments.

In summary, RAOPs demonstrate significant promise for next-generation energy storage systems due to their tunable redox properties, lightweight nature, and structural flexibility. Their successful implementation in lithium-ion, sodium-ion, and proton batteries, as well as in supercapacitors, has shown that RAOPs can serve as viable alternatives to conventional inorganic materials. Molecular engineering strategies such as conjugated backbones, hybrid nanostructures, and redox-active functional groups have substantially improved performance metrics like capacity, cycling stability, and rate capability. However, challenges such as poor electronic conductivity, limited long-term stability, and solubility in electrolytes still limit large-scale adoption. Future research should focus on developing highly conductive RAOPs, enhancing stability through crosslinking and composite design, and exploring scalable green synthesis methods. By addressing these hurdles, RAOPs are poised to play a transformative role in sustainable and flexible energy storage technologies.

Organic photovoltaics

Redox-active organic polymers have emerged as highly promising materials for advancing OPVs, owing to their tunable electronic properties and capacity to undergo reversible oxidation-reduction reactions [144]. These polymers, capable of donating and accepting electrons via redox-active sites such as quinones, phenazines, and TEMPO radicals, play a pivotal role in the active layers of OPV devices by facilitating charge generation, transport, and separation [145]. Their integration into bulk heterojunction (BHJ) architectures, typically as electron donors paired with electron-accepting materials like fullerene derivatives, enhances exciton dissociation and charge carrier generation—key processes that elevate the power conversion efficiency (PCE) of solar cells [146]. The versatility of these polymers, stemming from their solution-processability and mechanical flexibility, also makes them suitable for scalable, cost-effective fabrication and applications in wearable electronics and large-area photovoltaics [147].

Recent advancements in the design of redox-active organic polymers have focused on optimizing their optical and electronic properties to further improve OPV performance. Donor-acceptor (D-A) copolymers, which combine electron-rich and electron-deficient units, have demonstrated significant enhancements in light absorption and charge transport by allowing precise tuning of bandgap and energy levels. This molecular engineering reduces energy losses and boosts light-harvesting efficiency. For instance, Al-Isaee et al. [148] conducted a study on the synthesis and characterization of a new series of 2,6-linked anthracene-benzothiadiazole-based conjugated polymers for application in organic solar cells. The researchers aimed to develop

medium bandgap polymers with alternating donor-acceptor structures, utilizing anthracene as a weak electron donor and benzothiadiazole (BT) as an electron acceptor. The polymers were designed with various substituents at the 9,10-positions of the anthracene units to enhance electronic conjugation and planarity, which are critical for improving charge mobility and light absorption. The study explored the optical, electrochemical, and thermal properties of the synthesized polymers, as well as their potential in photovoltaic applications. The polymers exhibited multiple absorption bands and relatively low optical bandgaps, indicating their suitability for solar energy harvesting. However, the polymers' low solubility in common organic solvents posed challenges for device fabrication, limiting their photovoltaic performance. The work highlights the importance of structural modifications in conjugated polymers to optimize their properties for solar cell applications, while also pointing out the need for further improvements in solubility and processability to enhance device efficiency.

Beyond improving efficiency, the incorporation of redox-active moieties like TEMPO radicals or quinones has been shown to enhance the stability of charge transport pathways, reducing recombination losses and extending device longevity. Mikie et al. [149] provided further evidence of this potential in their investigation of ester-functionalized quinoxaline (QE)-based π -conjugated polymers for OPVs. By designing polymers PTQE-C8 and PTQE-EH with electron-deficient QE units, they achieved low-lying HOMO energy levels and narrow optical bandgaps—crucial for high photovoltaic performance. The polymer PTQE-C8, with its linear alkyl chains, outperformed its branched counterpart PTQE-EH, displaying higher crystallinity, extended π -conjugation, and superior charge transport in both fullerene and non-fullerene acceptor-based OPVs. Notably, these QE-based polymers also exhibited n-type semiconducting behavior, broadening their applicability and highlighting the versatility of redox-active building blocks in tailoring polymer properties for efficient solar cells.

While these developments are promising, challenges persist in translating redox-active organic polymers into widespread OPV applications. Issues such as long-term environmental stability, scalability of synthesis, and compatibility with large-scale manufacturing require further investigation. A deeper understanding of structure-property relationships is also essential for continued optimization. Insights from related fields, such as energy storage, offer additional perspectives on overcoming these hurdles. For example, Prasad et al. [150] investigated the photostability of various donor and acceptor materials used in organic solar cells (OSCs), focusing on the impact of molecular structure on their degradation under simulated sunlight. The study examined the photodegradation of donor polymer PBDB-T, small molecule acceptor Y5, and copolymer acceptors PFS-Y5 and PYT, using a combination of spectroscopic techniques, including UV-Vis absorption, FTIR, XPS, and UPS. The authors found that Y5 and PYT exhibited superior photostability, with minimal changes in their absorption spectra and no evidence of oxidation products after prolonged light exposure. In contrast, PBDB-T and PFS-Y5 underwent rapid photobleaching and photo-oxidation, forming new carbonyl and sulfur oxide groups. The study highlighted the role of the benzodithiophene-thiophene moiety in accelerating photodegradation, as its replacement with a thiophene unit in PYT significantly enhanced stability. The findings suggest that molecular design, particularly the choice of co-monomer, plays

a critical role in the intrinsic photostability of OSC materials, providing valuable insights for developing more durable organic photovoltaic devices.

In conclusion, the continuous evolution of redox-active organic polymers holds transformative potential for organic photovoltaics. By leveraging advanced synthetic approaches and redox-active functionalities, as demonstrated in the works of Smida et al., Mikić et al., and Jin et al., researchers are steadily overcoming existing limitations. These efforts pave the way for more efficient, stable, and cost-effective solar energy technologies, contributing to the broader adoption of sustainable energy solutions. However, addressing remaining challenges through interdisciplinary insights and refined design strategies will be critical to realizing their full impact in revolutionizing the field of OPVs. Building on the advancements in redox-active organic polymers, Smida et al. [151] synthesized a novel soluble semi-conducting polymer, AnFL, incorporating alternating anthracene and fluorescein photoactive units via Williamson polycondensation, aiming to enhance organic photovoltaic performance. The polymer's structural, optical, and electrical properties were comprehensively characterized using techniques such as NMR, FTIR, UV-visible absorption, and photoluminescence spectroscopies, alongside cyclic voltammetry and current-voltage measurements. The study revealed significant fluorescence resonance energy transfer (FRET) and photoinduced electron transfer (PET) from the anthracene core to the fluorescein unit, which are crucial for improving photovoltaic efficiency. Theoretical calculations using density functional theory (DFT) and time-dependent DFT (TD-DFT) supported the experimental findings, providing insights into the polymer's electronic structure and charge transfer mechanisms. The fabricated single-layer [ITO/AnFL/Al] device exhibited typical diode behavior with a space-charge-limited current conduction mechanism, demonstrating promising charge carrier mobility. Additionally, the AnFL: PCBM nanocomposite showed efficient interfacial charge transfer, suggesting potential for high photovoltaic performance. Overall, the work highlights the potential of AnFL as a functional polymer for organic photovoltaics, with ongoing research aimed at optimizing its performance in various nanocomposite structures. In a similar vein, Yang et al. [152] conducted a comprehensive study comparing the synthesis and performance of π -conjugated polymers for organic solar cells (OSCs) using two distinct polymerization methods: direct C–H arylation polymerization (DARp) and Stille coupling. The primary goal of the research was to evaluate whether DARp, an emerging and more atom-economical synthetic approach, could produce polymers with photovoltaic performance comparable to those synthesized via the traditional Stille method, which is known for its efficiency but also for generating toxic tin waste. The authors designed and synthesized a low-defect polymer, PBDTF-TPD, using both methods and systematically compared their optoelectronic properties, photovoltaic performance, and film morphology (Fig. 3A). Their findings revealed that the DARp-derived polymer exhibited similar structural regularity, optical properties, and device efficiency to its Stille-derived counterpart, achieving a power conversion efficiency (PCE) close to 9%. This work highlights the potential of DARp as a viable, environmentally friendly alternative to Stille coupling for the synthesis of high-performance π -conjugated polymers, provided that the monomers are carefully selected to minimize branching defects. The study contributes to the ongoing efforts to develop sustainable synthetic routes for organic photovoltaic materials,

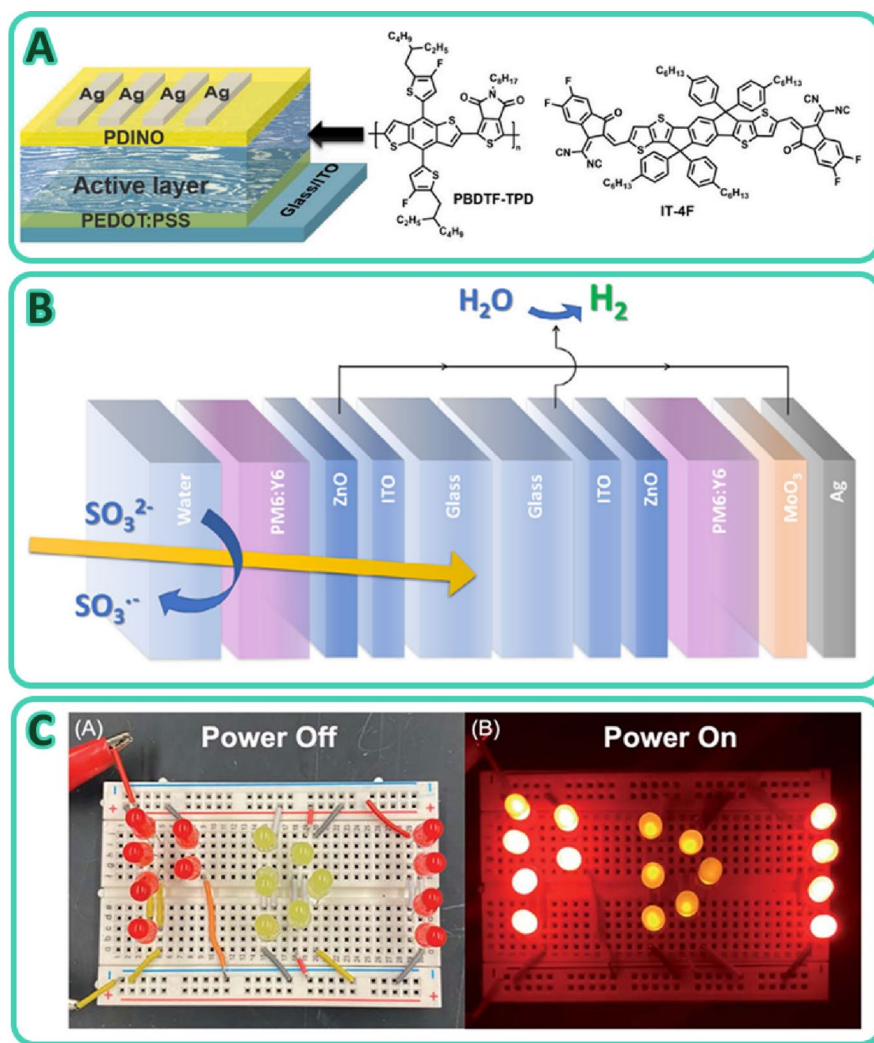


Fig. 3 (A) Schematic representation of conventional BHJ device architecture with respective (D/A) molecular structures of PBDTF-TPD and IT-4 F. Reprinted with permission from [152]. (B) Schematic representation of the modeled structure. Reprinted from [153]. (C) Practical demonstration of TAPPy-PDI electrode coin cells: (A) power off and (B) power on status of the LED array. Reproduced from Ref [157]. from the Royal Society of Chemistry

paving the way for future advancements in OSC technology. Expanding the scope of organic photovoltaic materials, Bernal-TeXca et al. [153] explored the potential of organic photo-absorbing blends, particularly PM6:Y6, to overcome the limitations of metal oxide photoanodes in tandem photovoltaic-photoelectrochemical (PV-PEC) structures for solar fuel production. The study focused on the computational design and experimental fabrication of high-throughput organic photoelectrochemical cells (OPECs) and OPV tandem structures. By systematically testing various organic blends, including PTB7-Th: FOIC, PTB7-Th: O6T-4 F, PM6:Y6, and PM6:FM, the

authors identified PM6:Y6 as the most promising candidate due to its exceptional charge separation, low recombination losses, and high photocurrent generation. The research demonstrated the successful implementation of a double PM6:Y6 photoanode/photovoltaic tandem structure, achieving significant photocurrents at bias-free potentials, which is crucial for efficient solar synthetic fuel production (Fig. 3B). The work highlights the tunability of organic materials, their solution-based fabrication, and their potential to surpass the performance of traditional metal oxide photoanodes, paving the way for further exploration of organic-based systems in sustainable energy technologies. Further advancing the field, Balci et al. [154] explored the synthesis, characterization, and application of novel non-fullerene acceptors (NFAs) based on carbazole, fluorene, and silafluorene derivatives for OPV devices. The study aimed to address the limitations of fullerene-based acceptors, such as high costs and sensitivity to atmospheric conditions, by designing NFAs with tunable optoelectronic properties and enhanced light absorption capabilities. The researchers synthesized three NFAs—CBR, FSBM, and SFBM—through Pd-catalyzed Suzuki cross-coupling and Knoevenagel condensation reactions, focusing on the impact of atomic substitutions (nitrogen and silicon) on the electronic properties of the fluorene-based core. Electrochemical and optical analyses revealed that the electron-donating nature of the carbazole and silafluorene cores contributed to improved push-pull characteristics, while the electron-deficient fluorene core reduced device performance. The NFAs were integrated into bulk heterojunction (BHJ) OPVs alongside donor polymers P3HT and PTB7-Th, with the inverted device structure yielding the best performance. Morphological studies using AFM and TEM highlighted the importance of film homogeneity and phase separation in enhancing device efficiency. The work underscores the potential of NFAs in advancing OPV technology by offering a viable alternative to fullerene-based systems, with a focus on optimizing molecular design for better performance and stability. Complementing these efforts, Xiong et al. [155] explored the strategic incorporation of fluorine atoms into the alkyl side chains of polymer donors to enhance the performance of OSCs. The study focused on synthesizing four polymers (PTF0, PTF1, PTF2, and PTF3) with varying degrees of fluorine functionalization, aiming to optimize their optoelectronic properties and intermolecular interactions. The researchers demonstrated that precise control over fluorine content, particularly in PTF1 with a single fluorine atom, significantly improved energy level alignment, aggregation, and crystallinity, leading to efficient charge generation and transport. This optimization resulted in binary and ternary OSCs achieving notable power conversion efficiencies. The work highlights the potential of fluorine functionalization in polymer donors as a viable strategy for advancing OSC performance, offering insights into the design of next-generation materials for high-efficiency solar cells. Shifting focus to energy storage applications, Kivala et al. [156] explored the synthesis and characterization of novel microporous polymers based on spirofluorene-bridged N-heterotriangulenes (N-HTAs) for potential applications as organic cathode materials in lithium-ion batteries. The study focused on the development of redox-active, photoluminescent porous polymers through Yamamoto polymerization of selectively brominated N-HTA monomers. The resulting polymers exhibited high thermal stability, significant surface areas, and reversible redox activity, making them promising candidates for energy storage applications. The authors demonstrated that the poly-

mers retained the electrochemical properties of the N-HTA units, with specific capacities and cycling stability comparable to existing organic cathode materials. The work highlights the potential of spirofluorene-bridged N-HTAs as versatile building blocks for functional porous polymers, offering insights into the design of materials with enhanced porosity and redox activity for future battery technologies. Similarly, Jin et al. [157] developed a novel approach to synthesize π -conjugated redox-active two-dimensional polymers (RA-2DPs) for use as organic cathode materials in lithium batteries. The authors designed a new polycyclic aromatic redox-active building block, PDI-DA, and utilized a modular synthesis strategy to create two RA-2DPs, TAPPy-PDI and TAPB-PDI, through imine-condensation followed by quantitative alkyl chain removal. These materials exhibited high electrical conductivity and ion diffusion rates due to their in-plane conjugation and permanent porosity, addressing the limitations of previous non-conjugated RA-2DPs (Fig. 3C). The study demonstrated that both RA-2DPs functioned effectively as cathode materials, showing good rate performance and excellent cycling stability. Additionally, the authors showcased the practical application of TAPPy-PDI by powering a series of LED lights, highlighting the potential of these materials for high-performance organic energy storage devices. The work underscores the importance of structural design and synthetic strategies in developing advanced organic cathode materials for next-generation batteries.

Incorporation of RAOPs into donor–acceptor architectures and bulk heterojunction systems has significantly improved power conversion efficiencies and photostability. Recent advances in molecular design, including side-chain engineering, conjugation extension, and the introduction of redox-active moieties, have expanded the potential of RAOPs for both binary and tandem OPV systems. Despite these achievements, challenges such as long-term stability under environmental stress, limited solubility, and suboptimal energy-level alignment persist. Future efforts should prioritize the development of more stable redox-active polymers, non-toxic processing solvents, and scalable synthesis routes, alongside integrating computational design tools to accelerate material discovery. RAOPs hold strong potential for enabling cost-effective, flexible, and sustainable solar energy technologies.

Thermoelectric devices

Thermoelectric devices, which convert heat directly into electrical energy, have garnered significant attention in recent years due to their potential for sustainable energy harvesting, particularly in waste heat recovery applications [158]. Among the materials driving this innovation, redox-active organic polymers stand out for their reversible oxidation and reduction properties, offering a promising, eco-friendly alternative to traditional inorganic thermoelectric materials that often depend on scarce or toxic elements [159]. These polymers possess inherent advantages, such as flexibility, low thermal conductivity, and tunable chemical structures, which allow precise control over their thermoelectric properties—namely, electrical conductivity, Seebeck coefficient, and thermal conductivity [160]. The incorporation of redox-active groups, such as quinones, thiophenes, or tetrathiafulvalenes, enhances charge transport and carrier mobility within the polymer matrix, optimizing the power factor—a critical metric

defined as the product of the Seebeck coefficient squared and electrical conductivity. This balance is essential for improving the thermoelectric figure of merit (zT), which quantifies a material's efficiency in converting heat into electricity. Moreover, the reversible modulation of redox states in these materials enables dynamic adjustment of their thermoelectric performance in response to environmental conditions, further enhancing their practical utility [161].

A key advantage of organic polymers lies in their intrinsically low thermal conductivity, which minimizes heat loss and boosts energy conversion efficiency—a critical factor in thermoelectric applications. This property, combined with the ability to fine-tune electronic structures through redox-active groups, has positioned these materials as viable candidates for next-generation thermoelectric systems. For instance, Han et al. [162] explored the thermoelectric potential of a novel thiophene-free conjugated polymer, poly(2-(12,2'-biselenophen)-5-yl)-3,8-difluoro-5,10-bis(5-octylpentadecyl)-5,10-dihydronidolo[3,2-b]indole) (PIDF-BSe), designed to enhance molecular doping efficiency. Incorporating electron-donating indoloindole and biselenophene moieties, the study examined the effects of two dopants: an electron-accepting molecule (F4TCNQ) and a Lewis acid (AuCl_3). Through advanced spectroscopic techniques, including UV-Vis, FT-IR, Raman, and X-ray photoelectron spectroscopy (XPS), the researchers identified the selenium atoms in biselenophene and nitrogen atoms in indoloindole as primary interaction sites for dopants. Their findings revealed that while AuCl_3 doping yielded higher electrical conductivity due to increased charge carrier concentration, F4TCNQ doping achieved superior thermoelectric performance, with higher Seebeck coefficients and power factors. This case underscores the pivotal role of dopant selection in optimizing the delicate interplay between electrical conductivity and Seebeck coefficient, offering valuable insights into designing efficient doping strategies for organic thermoelectric materials. Recent advancements have further demonstrated the practical applicability of redox-active organic polymers in thermoelectric devices, particularly for low-temperature energy harvesting and flexible electronics. PEDOT, a widely studied polymer, exemplifies this potential when doped with redox-active counterions, achieving significant improvements in electrical conductivity and moderate Seebeck coefficients. Building on this, Lee et al. [163] investigated PEDOT derivatives synthesized with a three-armed conjugated cross-linker, 1,3,5-tri(2-thienyl)-benzene (TTB), to explore the impact of polymer chain architecture on thermoelectric performance. Using electrochemical polymerization, they produced a range of structures—linear PEDOT, two-dimensional PEDOT (2D PEDOT), PEDOT-co-TTB copolymers, and polymerized TTB (poly-TTB)—and varied TTB concentration to assess its effects. Their results showed that a small amount of TTB enhanced π - π stacking interactions, boosting electrical conductivity and power factor, whereas excessive TTB disrupted chain conjugation, diminishing performance. The optimized 2D PEDOT structures highlighted in this study offer particular promise for flexible thermoelectric devices, illustrating how strategic modifications to polymer architecture can enhance thermoelectric efficiency and processability for real-world applications, such as wearable energy harvesters powered by body heat.

Despite these advances, challenges persist in scaling up redox-active organic polymers for widespread thermoelectric use. Long-term stability under operational con-

ditions, reproducibility in large-scale synthesis, and the inherent trade-off between electrical conductivity and Seebeck coefficient remain significant hurdles. To address these issues, hybrid approaches combining organic polymers with inorganic nanostructures have emerged as a promising direction. For example, Lin et al. [164] investigated the thermoelectric performance of MOF/CNT hybrids, aiming to overcome the poor electrical conductivity and processability of MOFs alone. By integrating CNTs, which enhance electrical and mechanical properties, with MOFs like MIL-68(Al) and MIL-53(Al)—each with distinct pore geometries—the researchers achieved n-p type carrier transport inversion and optimized zT values for both n-type and p-type hybrids. The low thermal conductivity of MOFs, paired with the conductivity boost from CNTs, resulted in a flexible thermoelectric generator suitable for wearable and IoT applications. This study elucidates the critical relationship between framework structure, oxygen adsorption, and thermoelectric properties, providing a blueprint for designing high-performance hybrid materials that leverage the strengths of both organic and inorganic components.

Looking forward, the future of redox-active organic polymers in thermoelectric applications hinges on overcoming these challenges through innovative material design [165–167]. Research efforts are likely to focus on developing novel redox-active monomers, advanced doping techniques, and hybrid systems to further enhance zT and overall device performance [168]. Understanding the intricate interplay between molecular structure, redox activity, and thermoelectric properties will be paramount in creating materials that meet the demands of sustainable energy technologies. By addressing stability, scalability, and efficiency concerns, redox-active organic polymers could play a transformative role in thermoelectric systems, enabling the efficient harnessing of waste heat and ambient thermal energy to power a range of applications—from small electronics to large-scale energy recovery systems—thus contributing to a more sustainable energy landscape. In another study highlighting bio-inspired innovations, Feng et al. [169] developed a novel conductive nanofiber with a core-sheath structure, inspired by mussel-inspired polydopamine (PDA) chemistry, for multifunctional applications in thermoelectric materials and humidity sensing (Fig. 4A). The study focused on the in-situ polymerization of EDOT on the surface of tunicate sulfated nanocellulose (TSN) mediated by PDA, resulting in PEDOT: PDA: TSN (PPTSNT) nanofibers (Fig. 4B). The core-sheath structure leveraged the mechanical strength of TSN and the electrical conductivity of PEDOT, enabling the fabrication of flexible, self-supporting nanocomposite films. These films exhibited excellent thermoelectric properties, including low thermal conductivity and decent power factors, making them suitable for energy harvesting applications. Additionally, the humidity-responsive behavior of PPTSNT was explored, demonstrating its potential in real-time respiratory monitoring and non-contact sensing for human-computer interaction (Fig. 4C). The work highlights the integration of bio-inspired surface modification strategies to enhance the dispersion and interfacial compatibility of PEDOT with nanocellulose, expanding its applications in flexible electronics and sensing technologies. Further exploring the domain of conductive polymers, Perry et al. [170] demonstrated a novel template-free electrochemical method for depositing nanostructured poly-3-hexylthiophene (P3HT) films, which eliminates the need for surface pre-treatments or self-assembled monolayers typically required for even

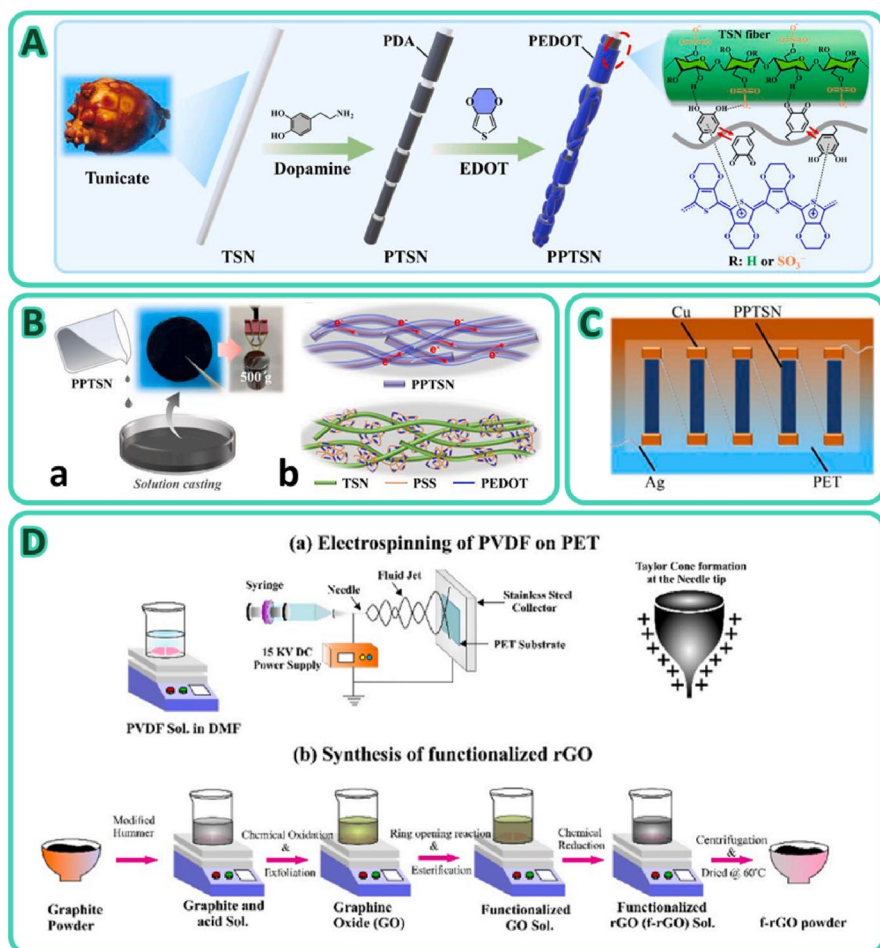


Fig. 4 (A) Schematic diagram for the preparation of PPTSN with core-sheath structure. Reprinted with permission from [169]. (B) (a) Schematic diagram of PPTSN nanofilms prepared by solution casting method (Inset shows PPTSN-2 nanofilm can withstand a weight of 500 g). (b) Schematic diagram showing the conductive paths of PPTSN and PEDOT: PSS-TSN. Reprinted with permission from [169]. (C) Schematic diagram of the PPTSN TEG. Reprinted with permission from [169]. (D) Schematic diagram of synthesis of (a) PVDF mat on PET via electrospinning method, and (b) f-rGO via Modified Hummer's process. Reprinted with permission from [171]

film formation. The authors employed an initial anodic pulse to create nucleation sites across the electrode surface, followed by cyclic voltammetry to grow highly conductive and nanostructured P3HT films. The resulting films exhibited interconnected wire-like structures and high conductivity without the need for post-treatment or chemical doping. This approach provides a straightforward route for producing conductive organic films suitable for applications such as thermoelectric generators, organic solar cells, and field-effect transistors. The study highlights the potential of this method for fabricating flexible power generation devices and suggests its applicability to other conductive polymers through the selection of appropriate mono-

mers. In line with the focus on flexible and multifunctional materials, Kundu et al. [171] developed a novel ternary nanocomposite-based ammonia sensor by depositing polyaniline (PANI), functionalized reduced graphene oxide (f-rGO), and copper-zinc ferrite ($\text{Cu}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$) on a flexible electrospun polyvinylidene difluoride (PVDF) mat. The primary goal of the study was to create a highly sensitive, portable, and flexible ammonia sensor capable of detecting trace-level ammonia at room temperature, while also exploring the thermoelectric properties of the composite material (Fig. 4D). The researchers optimized the electrospinning time to achieve a PVDF mat with ideal porosity and uniformity, which enhanced the gas sensitivity of the sensor. The nanocomposite exhibited excellent responsivity, fast response and recovery times, and high selectivity towards ammonia, even at low concentrations. Additionally, the study investigated the thermoelectric performance of the composite, revealing a significant figure of merit value, indicating its potential for energy harvesting applications. The work demonstrated the successful integration of PANI, f-rGO, and copper-zinc ferrite into a flexible sensor, highlighting its dual functionality as both a gas sensor and a thermoelectric material, thus contributing to advancements in environmental monitoring and energy conversion technologies.

Additionally, tackling the challenge of low electrical conductivity in MOFs, Olorunyomi et al. [172] explored the simultaneous enhancement of electrical conductivity and porosity in MOFs for thermoelectric applications (Fig. 5A). The study focused on addressing the inherent challenge of low electrical conductivity in MOFs, which limits their practical use in thermoelectric energy conversion. By employing PEDOT: PSS as a nucleating agent, the researchers synthesized $\text{Cu}_3(\text{BTC})_2$ (CuBTC) composites, labeled CPP-y, which exhibited significantly improved electrical conductivity while maintaining high porosity and low thermal conductivity. The composites demonstrated a remarkable balance between these properties, achieving high Seebeck coefficients and a notable zT for MOF-based materials. The work highlights the potential of MOF-based composites for thermoelectric applications, particularly in low-grade heat recovery, and opens avenues for further exploration in energy conversion technologies. Expanding on the application of advanced nanocomposite materials, Ji et al. [173] developed a multifunctional MXene/polyvinyl alcohol (PVA) hydrogel that serves as both a continuous ionic thermoelectric (i-TE) generator and a strain/temperature sensor (Fig. 5B). The study focuses on harnessing low-grade thermal energy through the thermoelectric effect, utilizing the thermo-diffusion of Cu^{2+} and Cl^- ions, along with redox reactions involving Cu/Cu^{2+} at electrode interfaces. The incorporation of MXene into the PVA hydrogel enhances ion diffusivity, leading to a high thermopower and rapid response times. The system achieves continuous current output by leveraging redox reactions at copper electrodes, making it suitable for long-term energy harvesting applications (Fig. 5C). Beyond energy generation, the hydrogel demonstrates multifunctionality as a sensor, capable of detecting mechanical strain via the piezoresistive effect and locating contact points or measuring temperature through the ionic thermoelectric effect. The work highlights the potential of MXene/PVA hydrogels in wearable electronics and low-power devices, bridging the gap between traditional electronic and ionic thermoelectric materials by offering both high voltage output and continuous energy supply. Finally, advancing the multifunctional properties of PEDOT: PSS, Tu et al. [174] explored

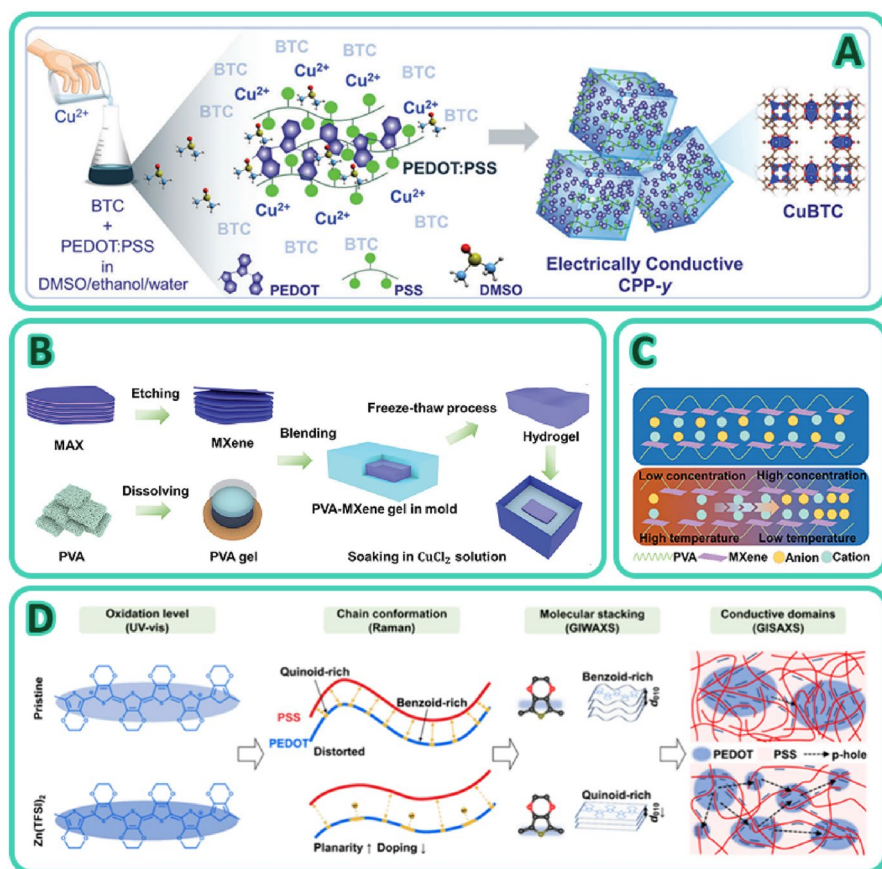


Fig. 5 (A) The precursors consisting of a fixed amount of Cu^{2+} salt and a changing ratio of PEDOT:PSS to BTC dissolved in ethanol: water: DMSO solvent system are mixed in a one-step reaction at room temperature. CPP-y formation occurs immediately. Reprinted with permission from [172]. (B) Schematic illustration of the preparation process of MXene/PVA hydrogel. Reprinted with permission from [173]. (C) Illustration of the mechanism of ionic thermoelectric effect. Reprinted with permission [173]. (D) Schematic diagram of multilength scale structure of the pristine and $\text{Zn}(\text{TFSI})_2$ -post-treated PEDOT:PSS films. Reprinted with permission from [174] Copyright 2024 American Chemical Society

the electrostatic tailoring of freestanding PEDOT:PSS films using a simple ionic salt post-treatment strategy to enhance their multifunctional properties for applications in thermoelectrics, hydrogels, and actuators. The study focused on optimizing the electrical conductivity and Seebeck coefficient of PEDOT:PSS films by modulating the intrinsic electrostatic interactions within the polymer matrix, particularly through the use of zinc di[bis(trifluoromethylsulfonyl)imide] ($\text{Zn}(\text{TFSI})_2$). The researchers demonstrated that this post-treatment not only improved the electrical properties but also endowed the films with superior water stability, mechanical flexibility, and humidity responsiveness (Fig. 5D). By employing multidimensional characterizations, including spectroscopic and microstructural analyses, the authors elucidated the mechanisms underlying these enhancements, such as changes in polymer con-

formation, doping levels, and intermolecular interactions. The resulting freestanding films were successfully integrated into various applications, including flexible thermoelectric generators, conductive hydrogels, and humidity-responsive actuators, showcasing their potential in next-generation wearable electronics, bioelectronics, and soft robotics. This work provides a comprehensive understanding of how ionic salt treatments can overcome the trade-offs typically associated with PEDOT: PSS, offering a versatile platform for multifunctional organic electronic devices.

RAOPs offer a compelling platform for thermoelectric applications due to their intrinsic low thermal conductivity, mechanical flexibility, and tunable redox behavior. Their performance can be finely adjusted through molecular doping, backbone engineering, and incorporation into hybrid composites with carbon-based or inorganic materials. Recent studies have shown promising zT , particularly for flexible and low-temperature applications such as wearable electronics and waste heat recovery. Nevertheless, trade-offs between electrical conductivity and Seebeck coefficient, as well as long-term stability and reproducibility, remain major bottlenecks. To overcome these issues, future work should focus on optimizing dopant–polymer interactions, developing redox-switchable architectures, and leveraging bioinspired or hierarchical design strategies. With continued innovation, RAOP-based thermoelectrics could play a pivotal role in decentralized energy generation and multifunctional smart systems.

Electrocatalysis for hydrogen evolution

RAOPs have recently emerged as a promising class of materials for catalyzing hydrogen evolution reactions (HER), drawing significant attention due to their distinct advantages over traditional inorganic catalysts [175]. Composed of abundant elements such as carbon, hydrogen, and nitrogen, RAOPs offer a sustainable and cost-effective alternative to expensive platinum-based catalysts [176–179]. Their tunable electronic properties enable researchers to modify their structure and redox activity, optimizing their catalytic performance for HER. This tunability stems from their capacity to undergo reversible electron transfer processes, a critical feature that facilitates the efficient transfer of electrons necessary for hydrogen generation. As the demand for renewable energy solutions intensifies, RAOPs are increasingly positioned as viable candidates for sustainable hydrogen production [180].

The design of RAOPs for HER applications often involves the strategic incorporation of functional groups or heteroatoms, such as nitrogen, sulfur, or oxygen, into the polymer backbone to enhance electrocatalytic performance. These modifications improve the material's conductivity, increase the density of active sites, and promote proton adsorption, all of which are essential for efficient HER catalysis [181]. Conjugated microporous polymers (CMPs) and covalent organic frameworks (COFs), for instance, stand out due to their high surface areas, well-defined pore structures, and ability to integrate redox-active sites. These structural attributes not only enhance the accessibility of active sites but also facilitate the transport of reactants and products, thereby boosting overall catalytic efficiency. A notable example of this approach is provided by Wang et al. [182], who explored the development of phosphonitrilic-bridged metal porphyrin porous covalent organic polymers (COPs) with a push-pull

motif. In their study, two COPs—NiTPP-PCTCOP and CoTPP-PCTCOP—were synthesized and evaluated for HER performance in alkaline media. The researchers found that NiTPP-PCTCOP outperformed its cobalt counterpart, achieving lower overpotentials and demonstrating long-term stability. This superior performance was attributed to its chemical composition, push-pull structure, and efficient charge transfer properties, underscoring the potential of combining phosphonitrilic chloride trimer with metal porphyrins to create effective HER electrocatalysts from abundant, cost-effective materials.

Recent advancements have further demonstrated that RAOPs can achieve HER activity comparable to some conventional inorganic catalysts, particularly when integrated with conductive substrates. Nitrogen-rich polymers, such as polyaniline derivatives and polypyrrole-based materials, have shown exceptional HER performance by stabilizing reaction intermediates and reducing the activation energy required for hydrogen production. The combination of RAOPs with materials like carbon nanotubes or graphene enhances their electrocatalytic efficiency by leveraging the high electrical conductivity and mechanical stability of these substrates. Wu et al. [183] conducted a study on the synthesis and characterization of three anion-regulated cobalt(II) coordination polymers (CPs) and explored their dual-functional applications in electrocatalytic HER and L-cysteine electrochemical sensing. The researchers designed and synthesized these CPs using a solvothermal method, incorporating different anions to regulate their structural properties. The resulting CPs exhibited distinct dimensional structures, including 2D layers and 1D hollow columns, which were confirmed through single-crystal X-ray analysis. The study aimed to address the growing interest in clean energy solutions by investigating the HER performance of these CPs in acidic media, demonstrating their potential as efficient electrocatalysts. Additionally, the CPs were evaluated for their electrochemical sensing capabilities, particularly for detecting L-cysteine, an important biomolecule. The findings revealed that the CPs exhibited low detection limits and high sensitivity, making them promising candidates for biosensing applications. Overall, the work highlights the significance of anion-regulated CPs in advancing both energy conversion technologies and electrochemical sensing platforms, providing valuable insights into the design of multifunctional molecular catalysts.

Beyond their direct application in HER, RAOPs and related materials have shown versatility in broader electrocatalytic contexts, offering insights into their mechanistic potential. Zou et al. [184] investigated the role of a Mo₂C-functionalized electrode interface in enhancing microbial cathode electrocatalysis, specifically for fumarate reduction by *Shewanella oneidensis* MR-1. While Mo₂C is recognized for its HER catalytic activity, this study constructed a nanostructured Mo₂C interface on a carbon cloth electrode and demonstrated significant improvements in current consumption density, even at potentials where HER does not occur. This enhancement was linked to the rapid electrochemical reaction of riboflavin at the Mo₂C interface, facilitating inward electron transfer via extracellular electron transfer mediators like c-type cytochromes. Although not directly focused on RAOPs, this work provides complementary insights into how functionalized interfaces can enhance electrocatalytic kinetics, offering potential strategies that could be adapted to improve RAOP-based systems.

Despite these promising developments, the practical application of RAOPs for HER faces several challenges. Long-term stability, scalability of synthesis, and precise control over polymer morphology and active site distribution remain significant hurdles [185–188]. For instance, while NiTPP-PCTCOP in Wang et al.'s study exhibited impressive stability, broader adoption requires consistent performance across diverse conditions [189]. Similarly, Patil et al.'s poly-PdNPC-KB composite, though effective, necessitates scalable production methods to compete with established catalysts. Overcoming these obstacles will demand innovative polymerization techniques, advanced characterization tools, and computational modeling to elucidate structure-activity relationships. As research progresses, RAOPs hold immense potential to transform sustainable hydrogen production, offering a cleaner, more affordable alternative to traditional catalysts and advancing the global pursuit of renewable energy solutions. Building on the advancements in RAOPs, Harada et al. [190] explored the potential of DNA-inspired polymers as metal-free electrocatalysts for the HER in acidic media. The study focused on synthesizing hydrogen-bonding conductive polymer catalysts (HCPCs) through oxidative chemical vapor deposition (oCVD) using nucleobases—adenine, guanine, cytosine, and thymine—as monomers. Among the polymers formed, poly-guanine (PG) exhibited significant catalytic activity for HER, while poly-adenine (PA) and poly-cytosine (PC) showed minimal or no activity (Fig. 6A). The researchers attributed PG's superior performance to its extensive hydrogen-bonding network, which facilitates the Volmer-Tafel mechanism, a key step in HER. The study also employed DFT calculations to analyze the electrostatic potential and hydrogen-bonding capabilities of the polymers, providing insights into the structural factors influencing catalytic efficiency. The work highlights the potential of bio-inspired, metal-free polymers as sustainable alternatives to platinum-based catalysts for hydrogen production, emphasizing the importance of hydrogen-bonding networks in designing efficient electrocatalysts. Shifting focus from RAOPs, Li et al. [191] explored the development of polyimide-based triazine-cored covalent organic frameworks (PIT-COFs) as supports for metal nanoparticles (Fe, Ni, Ru, and Rh) to enhance electrocatalytic HER in acidic media. The study aimed to address the challenges of low overpotential and high stability in HER electrocatalysts by designing COFs with large π -conjugated structures, which facilitate electron and proton transport (Fig. 6B). The researchers synthesized four PIT-COFs using melamine and various aromatic dimides, with the perylenediimide-based COF demonstrating optimal HER activity due to its extended conjugation. Rhodium nanoparticles supported on the perylenediimide-based COF exhibited exceptional HER performance, characterized by low overpotential and high stability after extensive cycling. The work highlights the potential of COFs as robust platforms for stabilizing metal nanoparticles, thereby improving catalytic efficiency and durability. This research provides valuable insights into the design of efficient and stable COF-based electrocatalysts for hydrogen production, contributing to the advancement of sustainable energy technologies. In another study, Yang et al. [192] developed a novel hybrid electrocatalyst, CoTHPP-PDCP@DWCNTs, for efficient HER by covalently functionalizing cobalt-porphyrin (CoTHPP) with chain-like polydichlorophosphazenes (PDCP) and integrating it onto double-walled carbon nanotubes (DWCNTs). The study aimed to address the limitations of traditional Pt-based catalysts by creating a cost-effective, non-noble metal

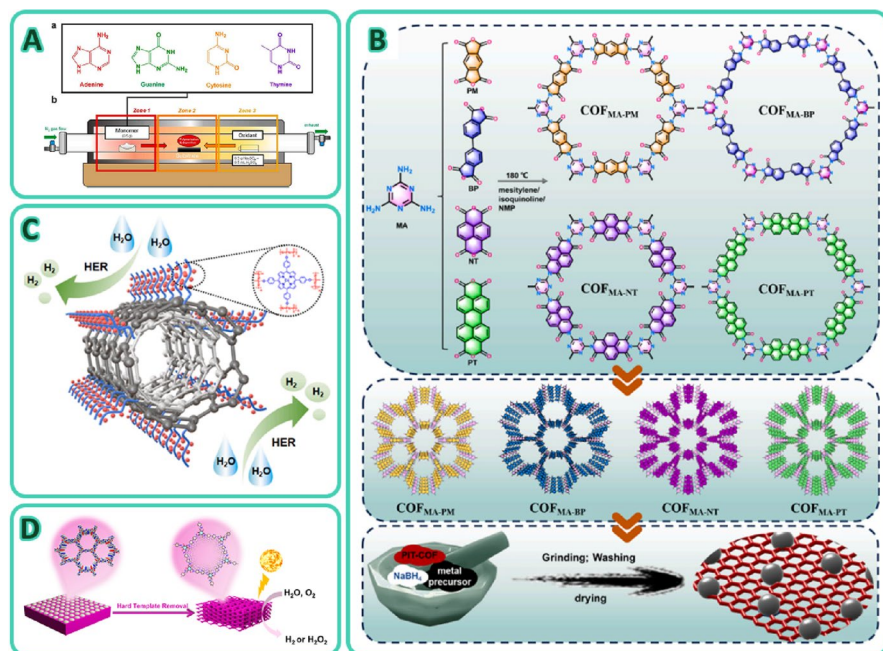


Fig. 6 (A) (a) DNA nucleobase monomers and (b) schematic image of the oCVD setup. Reprinted with permission from [190]. (B) Preparation of PIT-COFs and M@COFs. Reprinted with permission from [191]. (C) CoTHPP-PDCP@DWCNTs. Reprinted with permission from [192]. (D) Illustration of the Formation of Macro-TpBpy Employing Hard PSs as Template and Its Use for Photocatalytic HER and Hydrogen Peroxide Production. Reprinted with permission from [193] Copyright 2024 American Chemical Society

electrocatalyst with enhanced HER performance. The researchers demonstrated that the synergistic interactions between CoTHPP, PDCP, and DWCNTs significantly improved charge transfer, conductivity, and active site density, leading to reduced overpotentials and enhanced catalytic efficiency in both alkaline and acidic conditions (Fig. 6C). The work highlights the importance of covalent linkage and structural engineering in optimizing electrocatalysts for sustainable hydrogen production, offering insights into the design of next-generation high-performance electrocatalytic systems. Examining the role of hierarchical porous structures, Khalil et al. [193] explored the influence of hierarchical porous structures in COFs on photocatalytic performance, particularly focusing on HER and hydrogen peroxide (H₂O₂) production. The study aimed to address the limitations of microporous COFs in catalytic applications, where mass transport inefficiencies can hinder the accessibility of active sites. By introducing additional macropores into the COF structure, the researchers sought to enhance light penetration, improve reactant and product diffusion, and increase the overall surface area, thereby optimizing photocatalytic efficiency. The hierarchical porous COF, termed macro-TpBpy, was synthesized using polystyrene spheres as a hard template, resulting in a material with interconnected macropores alongside intrinsic microporosity (Fig. 6D). The authors demonstrated that the hierarchical structure significantly improved photocatalytic activity, with macro-TpBpy

outperforming its non-hierarchical counterpart in both HER and H_2O_2 production. The work underscores the importance of pore architecture in COFs for photocatalytic applications, highlighting how hierarchical porosity can enhance mass transport, light utilization, and overall catalytic performance. This study contributes to the growing body of research on COFs, emphasizing the need to consider structural design alongside chemical and electronic properties for optimizing photocatalytic systems.

Focusing on the structural design of electrocatalysts, Maeda et al. [194] investigated the synthesis and catalytic performance of bis(diimino)nickel coordination nanosheets (CONASHs) modified with triptycene moieties, focusing on their application in HER catalysis. The study aimed to address the limitations of conventional π -conjugated CONASHs, which often suffer from strong interlayer π - π interactions that shield active metal sites and reduce catalytic efficiency. By incorporating triptycene, a rigid, propeller-shaped molecule, the researchers designed a CONASH with perpendicular metal complex sites, which weakened interlayer interactions and facilitated exfoliation into thin films (Fig. 7A). The synthesized Ni-HATT

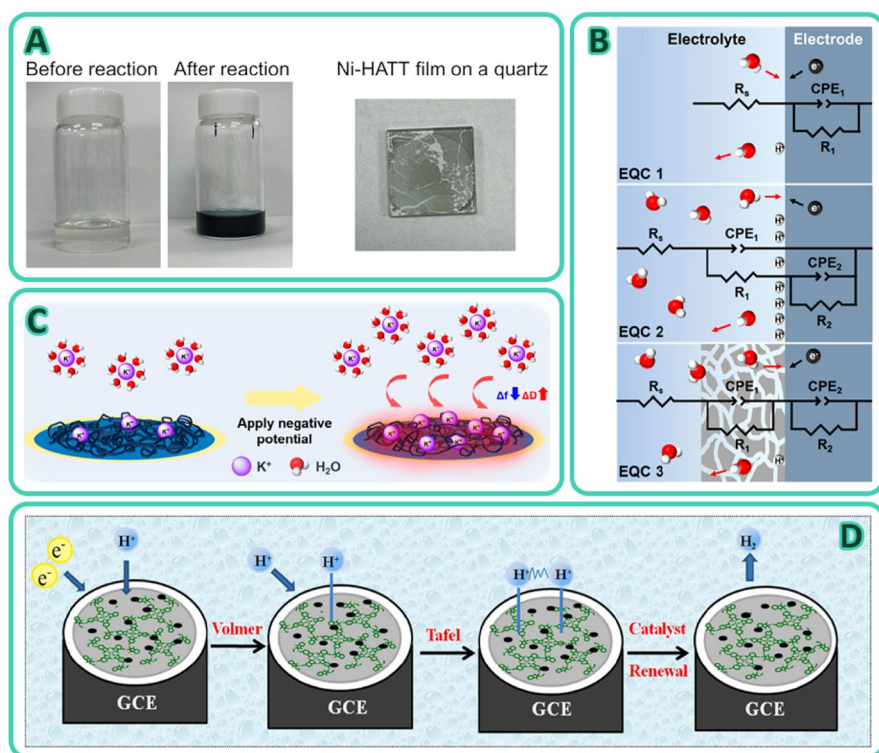


Fig. 7 (A) Photographs of a reaction vessel before and after the gas – liquid interfacial reaction and a Ni-HATT film on a quartz substrate. Reprinted with permission from [194] Copyright 2024 American Chemical Society. (B) Commonly used equivalent circuit models for HER and the illustration of electrode surface during HER process. Reprinted with permission from [195] Copyright 2024 American Chemical Society. (C) Illustration of the QCM sensor surface during EQCM-D measurements. Reprinted with permission from [195] Copyright 2024 American Chemical Society. (D) Volmer-Tafel mechanism for poly-PdNPc + KB. Reprinted with permission from [196]

nanosheets were characterized using various techniques, including Raman spectroscopy, X-ray photoelectron spectroscopy, and atomic force microscopy, confirming their structural integrity and enhanced accessibility to active sites. The Ni-HATT-modified electrodes demonstrated superior HER performance compared to conventional CONASHs, attributed to the exposed axial positions of the metal complexes, which allowed easier proton access. This work highlights the potential of triptycene-modified CONASHs as efficient electrocatalysts and underscores the importance of structural design in optimizing the performance of two-dimensional materials for energy applications. Furthermore, Luo et al. [195] investigated the enhancement of alkaline HER efficiency by modulating surface cation concentration through the molecular design of ethylene glycol (EG)-functionalized PEDOT derivatives. The study focused on synthesizing and characterizing three types of EG-functionalized EDOT monomers, which were electropolymerized on nickel foam (NF) electrodes to evaluate their HER performance (Fig. 7B). The authors demonstrated that the EG moieties in the PEDOT derivatives could induce a locally concentrated ion environment near the electrode surface, facilitating water dissociation and improving HER kinetics. Electrochemical impedance spectroscopy (EIS), electrochemical quartz crystal microbalance with dissipation (EQCM-D), and XPS analyses were employed to elucidate the mechanism behind the enhanced HER performance, revealing that the increased surface cation concentration under negative polarization played a crucial role (Fig. 7C). The study also highlighted the importance of molecular structure and EG chain length in determining the HER efficiency and stability of the electrodes. Overall, the work underscores the potential of rational molecular design in modifying electrode surfaces to achieve improved electrocatalytic performance for sustainable energy applications. Lastly, Patil et al. [196] explored the development of a cost-effective and stable electrocatalyst for the HER by synthesizing a polymeric palladium phthalocyanine (poly-PdNPC) hybridized with Ketjen Black (KB). The study aimed to address the limitations of platinum-based catalysts, such as high cost and scarcity, by leveraging palladium, which shares similar properties with platinum but at a lower cost. The researchers optimized the ratio of poly-PdNPC to KB to enhance electrochemical activity, achieving a hybrid material with superior HER performance in acidic media. The hybrid catalyst demonstrated enhanced conductivity, increased active surface area, and improved charge transfer kinetics, attributed to the synergistic effects of π - π interactions and electronic density modulation (Fig. 7D). The study also highlighted the importance of tuning the d-band center to optimize hydrogen adsorption and desorption, which are critical for efficient HER. Overall, the work presents a promising alternative to traditional HER catalysts, offering a sustainable and scalable solution for hydrogen production, which is crucial for advancing renewable energy technologies.

RAOPs have demonstrated substantial potential as electrocatalysts for HER, offering sustainable, metal-free, or low-metal alternatives to conventional catalysts. Their tunable redox sites, structural diversity, and ability to incorporate heteroatoms or coordinate with metal complexes make them highly adaptable for catalytic design. Advances in porous architectures, conjugated frameworks, and hybrid composites have enhanced electron transport, active site accessibility, and catalytic efficiency in both acidic and alkaline media. Despite this progress, challenges such as limited

long-term stability, low intrinsic conductivity, and insufficient mechanistic understanding still constrain their practical deployment. Future research should emphasize rational molecular design, operando characterization, and scalable synthesis techniques to optimize catalytic performance and durability. With continued development, RAOP-based catalysts may significantly contribute to clean hydrogen production and broader renewable energy platforms.

Carbon dioxide reduction

RAOPs have emerged as promising materials for CO₂ reduction, driven by their ability to undergo reversible electron transfer processes and convert CO₂ into valuable products such as carbon monoxide, formic acid, methanol, and even multi-carbon compounds like CH₄ and C₂H₄ [197]. These materials stand out due to their customizable electronic properties, including conjugated backbones that facilitate charge delocalization, enabling efficient activation of CO₂ through electron transfer from redox-active sites [198]. Compared to traditional inorganic catalysts, RAOPs offer advantages in cost, sustainability, and environmental friendliness, positioning them as viable candidates for advancing CO₂ mitigation efforts. This versatility and performance are exemplified in recent studies, such as those by Tang et al. [199], who explored donor- π -acceptor (D- π -A) CPs for photocatalytic CO₂ reduction under visible light. Their work focused on designing CPs with cyclotriphosphonitrile as the donor unit, benzene as the acceptor, and alkyne groups as π -bridges. By synthesizing three CPs with varying numbers of alkyne bridges, they found that the polymer with three π -bridges (TTB-PCT) achieved superior photocatalytic activity, yielding high amounts of CH₄ and C₂H₄ while maintaining excellent cycling stability. This study underscores how modulating the π -bridge structure enhances charge separation and stabilizes intermediates like $^*\text{CH}_2$ for C-C coupling, providing a foundation for designing efficient metal-free photocatalysts.

The mechanism of CO₂ reduction in RAOPs typically begins with CO₂ adsorption onto the polymer surface, followed by activation via electron transfer facilitated by redox-active functional groups such as quinones, imides, or conjugated aromatic systems. For example, quinone-based polymers can donate electrons to CO₂, producing carbon monoxide or other useful products. The ability to tailor RAOPs' properties—such as porosity, surface area, and redox activity—optimizes CO₂ adsorption and catalytic efficiency, allowing these materials to be adapted for specific reduction pathways. Recent advancements in synthesis techniques, including electrochemical polymerization and template-assisted methods, have further enhanced RAOPs' conductivity, stability, and selectivity. This is evident in the work of Yang et al. [200], who developed a novel polymeric carbon nitride (PCN) material, Brown PCN (Bro-PCN), to improve photocatalytic CO₂ reduction. By incorporating oxygen into the PCN structure through co-polymerization of glycolic acid and urea, they created Bro-PCN with a broad spectral response and sub-gap impurity states that improved light absorption, charge separation, and active site availability. Experimental and DFT analyses revealed that these modifications significantly boosted photocatalytic activity, particularly in the near-infrared region, highlighting the value of electronic structure optimization in RAOP design.

Beyond standalone polymers, integrating RAOPs with conductive substrates or nanomaterials has proven effective in enhancing charge transport and catalytic efficiency. A notable example is provided by Yang et al. [201], who investigated a dual-homojunction photocatalyst combining cyano/cyanamide-modified carbon nitride (CN-TH) with a 1T/2H-MoSe₂ homojunction. This innovative design tackled the high dissociation energy of CO₂ and sluggish charge transfer dynamics by leveraging the CN-TH homojunction for intralayer charge separation and the MoSe₂ homojunction for interlayer charge transport. The resulting tandem multi-step charge transfer pathway significantly improved CO₂ reduction efficiency, demonstrating the potential of hybrid RAOP systems. Such advancements underscore how nitrogen-rich polymers (e.g., triazine or porphyrin-based systems) or composite structures can enhance CO₂ affinity and catalytic performance under mild conditions, bolstering the commercial prospects of RAOPs as durable electrocatalysts. Despite these promising developments, challenges persist in translating RAOPs to large-scale CO₂ reduction applications. Long-term stability, scalability, and competition from HER remain significant hurdles [202]. To address these, future research must refine RAOPs' molecular design to improve selectivity and durability while integrating them into electrochemical or photochemical systems for enhanced efficiency. Techno-economic analyses and life cycle assessments will also be critical to assess their industrial feasibility and sustainability. The collective insights from studies like those by Tang et al. and Yang et al. illustrate the transformative potential of RAOPs [203, 204]. By combining tailored synthesis, strategic structural modifications, and hybrid system integration, RAOPs could play a pivotal role in global CO₂ mitigation, offering a sustainable pathway to convert a greenhouse gas into valuable chemical resources. Building on the fundamental understanding of RAOPs, Fera et al. [205] explored the development of a novel bioelectrocatalytic system for the efficient reduction of CO₂ to formate, addressing the limitations of traditional NADH-dependent enzymatic systems. The study focused on integrating *Candida boidinii* formate dehydrogenase (cbFDH) with an amino-viologen redox polymer (NH₂Et-PVI) to create a biocathode that eliminates the need for exogenous NADH, which is costly and requires continuous regeneration (Fig. 8A). The authors demonstrated that this system achieved high faradaic efficiency and significantly increased formate production compared to conventional NADH-dependent methods. By immobilizing the enzyme and redox polymer on a carbon cloth electrode, the system enabled continuous redox cycling, simplifying the reaction environment to only CO₂ and formate. This approach not only enhanced the thermodynamic feasibility of the reaction but also minimized byproduct formation, offering a scalable and cost-effective solution for sustainable CO₂ utilization. The work highlights the potential of artificial cofactors in advancing bioelectrochemical processes for industrial-scale carbon conversion. In parallel, Wang et al. [206] investigated the development of a heterogeneous carbon-nitrogen polymer heterojunction for enhanced photocatalytic CO₂ conversion. The study focused on constructing a donor-acceptor system by embedding electrophilic hexaazatriphenylene (HAT) into the heptazine framework of graphitic carbon nitride (g-C₃N₄) through in-plane fusion, forming a g-C₃N₄-C₂N composite. The authors demonstrated that the integration of C₂N into g-C₃N₄ significantly improved the separation of photoinduced electron-hole pairs, leading to enhanced photocatalytic activity (Fig. 8B). Furthermore, oxy-

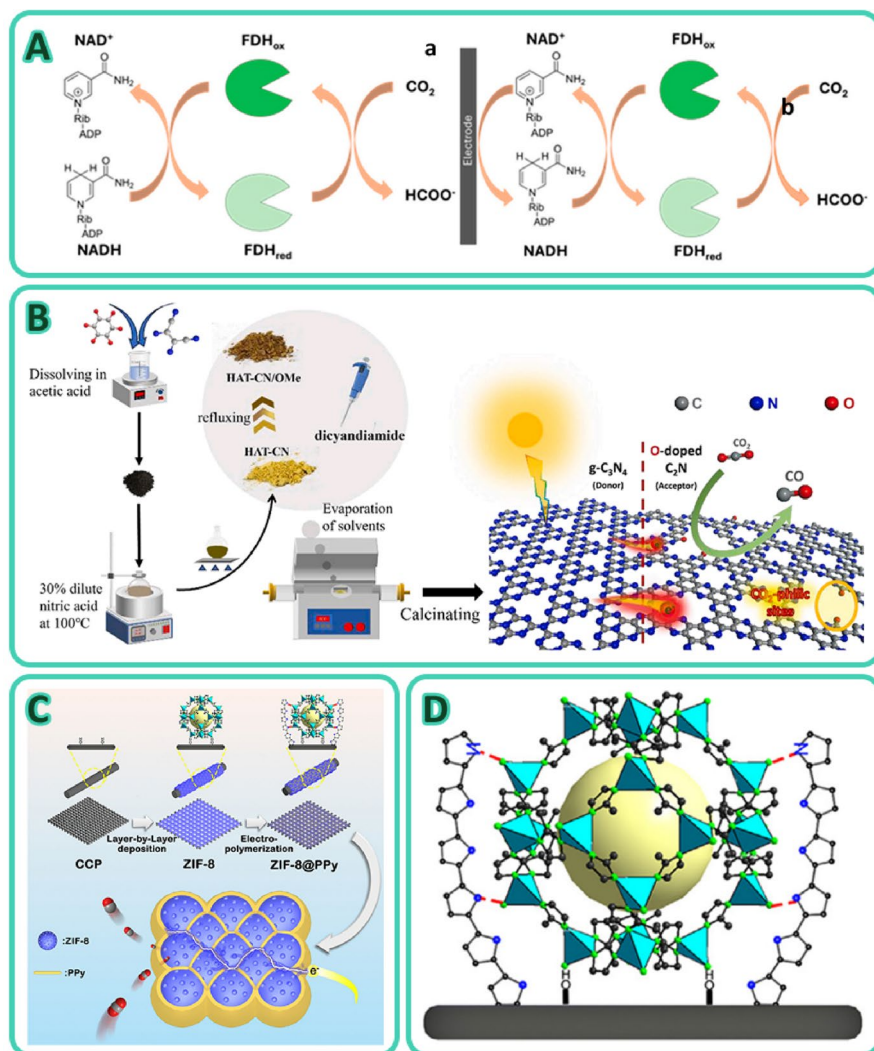


Fig. 8 (A) (a) Conventional biocatalysis with NADH, (b) direct electrochemical regeneration of NADH. Reprinted with permission from [205]. (B) Illustration for the synthesis of CC-O. Reprinted with permission from [206]. (C) Schematic illustration of the layer-by-layer deposition of binder-free, ZIF-8 nanofilm electrocatalysts on conductive carbonfiber paper and their electrochemical grafting of conductive PPY chains for electrocatalytic carbon dioxide reduction to carbon monoxide. (D) Schematic illustration of the grafting of PPY to ZIF-8 through the coordinated interaction between the pyrrolic N atoms of PPY and the Zn ions of ZIF-8. Reprinted with permission from [207]

gen doping was introduced to optimize CO₂ adsorption, resulting in a polymer with improved CO₂ photoconversion efficiency. The study revealed that the oxygen-doped polymer exhibited superior CO₂ reduction performance without the need for co-catalysts or sacrificial agents, achieving high selectivity for CO production. The work highlights the potential of combining donor-acceptor systems and oxygen doping in designing efficient metal-free photocatalysts for artificial photosynthesis, contribut-

ing to the broader goal of utilizing solar energy for carbon neutralization. Similarly, Li et al. [207] developed a binder-free, well-integrated MOF-based nanofilm electrocatalyst for the highly efficient and selective reduction of CO₂. The authors addressed the inherent limitations of MOFs, such as poor electrical conductivity, by grafting conductive polypyrrole (PPy) chains into a surface-mounted MOF (ZIF-8) nanofilm using a combination of layer-by-layer assembly and in-situ electrochemical polymerization techniques (Fig. 8C). This approach not only enhanced electron transfer but also optimized the electronic structure of the electrocatalyst, leading to improved utilization of active sites. The resulting ZIF-8@PPy nanofilm demonstrated exceptional electrocatalytic performance for CO₂ reduction to CO, achieving high selectivity and current density (Fig. 8D). The study highlights the potential of this strategy for designing MOF-based electrocatalysts with enhanced efficiency and selectivity for various electrochemical applications, contributing to the advancement of sustainable CO₂ conversion technologies.

Complementing these findings, Choi et al. [208] provide a comprehensive review of electrochemical direct CO₂ capture (EDCC) technology, focusing on the use of redox-active organic molecules as a promising approach to achieving carbon neutrality (Fig. 9A). The authors discuss the limitations of conventional carbon capture methods, such as amine scrubbing, which are energy-intensive and environmentally taxing, and highlight EDCC as a more sustainable alternative (Fig. 9B). The review delves into the fundamental principles of EDCC, including the design of redox-active organic sorbents, electrolyte selection, and device configurations, emphasizing the importance of molecular engineering to optimize CO₂ binding affinity and electrochemical performance. The article also explores the potential applications of EDCC in various contexts, such as direct air capture (DAC) and direct ocean capture (DOC), and draws parallels with other electrochemical systems like alkali-ion batteries and redox-flow batteries to inform the design of high-performance EDCC systems (Fig. 9C). The authors conclude by outlining the challenges and future directions for EDCC, underscoring the need for further research to enhance the stability, efficiency, and scalability of these technologies to realize their full potential in mitigating climate change. Further emphasizing the innovation in the field, Zhang et al. [209] investigated the enhancement of photocatalytic CO₂ reduction through the strategic design of metal-ion-coordinated COFs. The study focused on synthesizing benzoxazole-based COFs with distinct cobalt coordination geometries, namely Co–N–O₂, Co–N–O₃, and Co–N₂–O₂, to modulate exciton dissociation and improve the efficiency of CO₂ conversion to carbon monoxide (CO). By precisely tuning the coordination environment of Co ions, the researchers optimized the local electronic structure of the COFs, which facilitated charge carrier migration, mitigated electron-hole recombination, and reduced the energy barrier of the rate-determining step (Fig. 9D). The results demonstrated that the Co-coordinated COFs, particularly BBO-COFBPY-Co, exhibited significantly enhanced CO production rates and selectivity, coupled with excellent stability. The work underscores the pivotal role of metal coordination microenvironments in regulating excitonic behavior and advancing photocatalytic CO₂ reduction, offering a promising strategy for addressing climate change through efficient carbon conversion. Lastly, Luangchaiyaporn et al. [210] investigated the facile preparation of stable NiII- and CoII-tetraaminophthalocyanine (NiTAPc

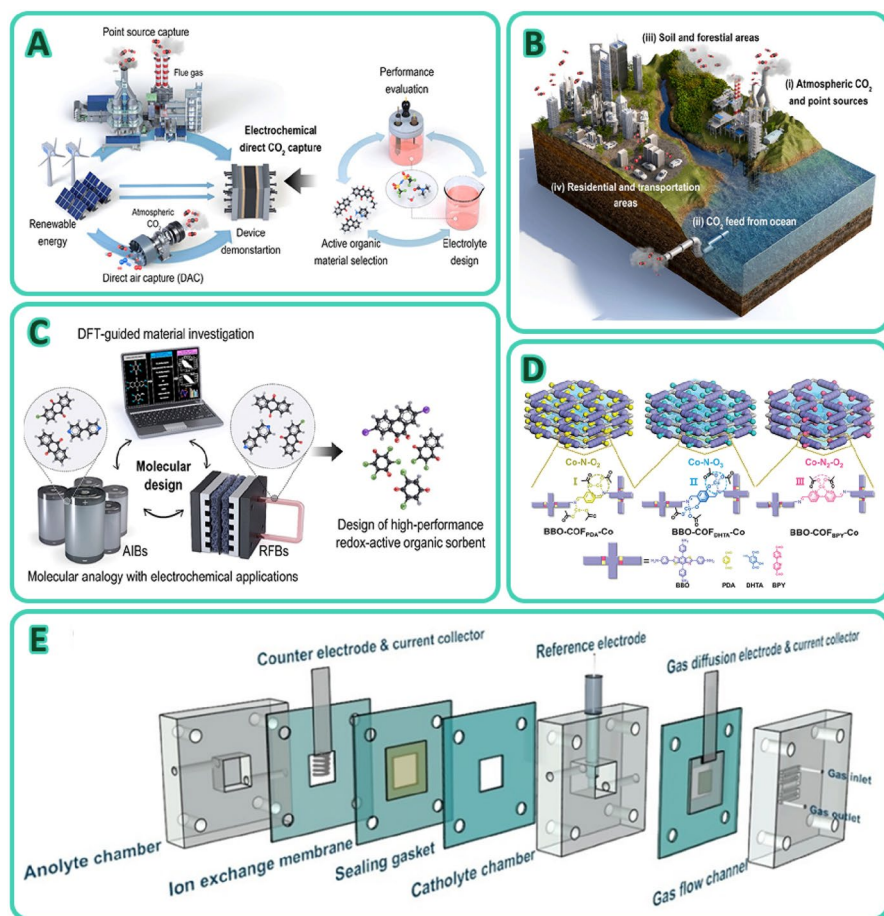


Fig. 9 (A) A schematic overview of EDCC technologies, including application areas and system constituents that influence the performance of EDCC. Reprinted with permission from [208]. (B) Possible applications of EDCC process with various CO₂ sources, containing (i) atmospheric CO₂, (ii) CO₂ feed from ocean, (iii) soil/forestal areas, and (iv) residential/transportation areas. Reprinted with permission from [208]. (C) Three representative strategies for the design of high-performance redox-active sorbent materials for ECC applications. Reprinted with permission from [208]. (D) Schematic illustration of the structure of BBO-COFs-Co. Reprinted with permission from [209]. (E) A schematic setup of flow cell components. Reprinted with permission from [210]

and CoTAPc) electropolymers for highly efficient heterogeneous electrochemical CO₂ reduction. The study focused on the development of polymer films, p-NiTAPc and p-CoTAPc, through electropolymerization of their corresponding monomers on carbon-based substrates without the use of binders or conducting additives, aiming to create efficient gas diffusion electrodes (GDEs) for scalable and selective CO₂-to-CO conversion (Fig. 9E). The polymers were characterized using various techniques, including UV-visible spectrophotometry, attenuated total reflection-Fourier transform infrared spectroscopy, scanning electron microscopy, energy dispersive X-ray spectroscopy, and cyclic voltammetry. The catalytic performance of these polymers

was evaluated in a flow electrolysis cell, where p-NiTAPc demonstrated high Faradaic efficiency and current density for CO production, while p-CoTAPc exhibited superior stability and productivity over extended periods. The study highlighted the potential of these electropolymers as effective catalysts for CO₂ reduction, offering a promising approach for large-scale applications in carbon capture and utilization technologies.

In the context of electrochemical CO₂ reduction, RAOPs present a sustainable and design-flexible alternative to traditional catalysts. Their ability to host redox-active sites, modulate charge distribution, and stabilize key intermediates makes them highly attractive for selective and energy-efficient CO₂ conversion. Incorporation of functionalities such as quinones, imines, or coordinated metal centers into RAOPs has enabled enhanced CO₂ adsorption and multi-electron transfer processes. Hybrid materials and porous architectures have further improved mass transport and catalytic activity. Nonetheless, achieving high selectivity, stability under long-term operation, and mechanistic clarity remain significant challenges. Future directions should include the development of site-specific functionalization strategies, integration with light-harvesting components for photo-electrocatalysis, and deeper insights from computational modeling and in situ spectroscopic techniques. RAOP-based systems hold promise for advancing CO₂ utilization technologies toward scalable and carbon-neutral energy cycles.

Catalysis in organic synthesis

RAOPs have emerged as a transformative class of materials in catalysis for organic synthesis, offering a sustainable and efficient alternative to traditional metal-based catalysts [211]. Their unique ability to undergo reversible redox reactions enables them to serve as electron donors or acceptors during catalytic cycles, a property critical for driving a variety of organic transformations, including oxidation, reduction, and C-H functionalization, often under mild conditions [212]. The polymeric nature of RAOPs enhances their catalytic efficiency by integrating multiple functional groups, which not only improves selectivity but also increases the number of active sites available for catalysis [213]. This versatility positions RAOPs as highly effective catalysts for a broad range of synthetic applications, such as cross-coupling reactions, polymerization, and asymmetric synthesis. Moreover, their flexibility and tunability allow them to mimic the functionality of transition metal catalysts while circumventing associated drawbacks, such as toxicity and environmental concerns, thereby addressing the growing demand for sustainable and recyclable catalytic systems in industrial applications [214].

A compelling example of RAOPs' potential in organic synthesis is illustrated by Moorthy et al. [215], who developed a novel ionic porous organic polymer (IPOP), CylTzTz-IPOP, as a heterogeneous photocatalyst for visible-light-mediated organic transformations. Synthesized via a nucleophilic substitution reaction between cyanuric chloride and 2,5-di(pyridin-4-yl)thiazolo[5,4-d]thiazole (BpyTzTz), this polymer exhibits excellent visible-light absorption, robust redox properties, and high porosity. The authors demonstrated its efficacy in catalyzing two distinct reactions: the synthesis of β -keto sulfoxides through tandem thiol-ene addition-oxidation and the

formation of benzothiophene diesters via thiol-yne addition-annulation. Both processes proceeded under mild conditions without additional additives, showcasing the catalyst's efficiency and environmental sustainability. Mechanistic studies revealed that single-electron transfer processes generated reactive intermediates, driving the formation of the desired products. Notably, CylTzTz-IPOP's recyclability was confirmed through multiple reuse cycles without significant loss of activity, underscoring its practical utility as a sustainable alternative to metal-based homogeneous catalysts.

Beyond their role in traditional organic synthesis, RAOPs demonstrate remarkable adaptability in specialized applications, such as energy storage systems. Zhang et al. [216] explored this potential in the context of rechargeable magnesium batteries (RMBs), addressing challenges related to Mg^{2+} insertion at the electrode-electrolyte interface. They developed a metal-organic polymer (MOP) cathode, Ni-TABQ (Ni-coordinated tetraminobenzoquinone), which leverages a 2D π -d conjugated structure to enhance surface catalytic activity. This design facilitates the dissociation of Mg_xCl_y clusters—a key barrier in RMBs—by reducing dissociation energy and enabling a four-electron redox process. The result is a significant improvement in battery capacity and cycling stability, highlighting the broader applicability of RAOPs' redox properties in electrocatalysis. This study exemplifies how the principles of RAOP catalysis can extend beyond organic synthesis to optimize the electrochemical performance of multivalent metal-ion batteries, paving the way for advancements in high-energy storage systems.

A notable recent advancement by Su et al. involved the development of a buta-1,3-diene-linked COF via multi-component synthesis, which showcased remarkable photocatalytic properties in heterogeneous organic transformations [217]. The extended π -conjugation provided by the diene linkages enabled efficient charge delocalization and visible-light harvesting, facilitating high redox activity under mild reaction conditions. This work not only demonstrates the potential of COF-based RAOPs in sustainable catalysis but also highlights the role of dynamic linkage chemistry in tuning both the redox and photophysical behavior of polymer frameworks for integrated light-driven processes.

In the realm of asymmetric synthesis, RAOPs offer promising opportunities for inducing enantioselectivity, a critical aspect of producing biologically active compounds. By incorporating chiral moieties into the polymer backbone, these materials can catalyze stereoselective reactions, such as asymmetric hydrogenation and epoxidation, with greater efficiency than small-molecule catalysts. The confined environment of the polymer matrix fosters cooperative effects between neighboring catalytic sites, further enhancing performance. Additionally, RAOPs' insolubility in organic solvents and their ability to be immobilized on solid supports make them ideal for continuous flow reactors, which are increasingly vital for large-scale synthesis in academic and industrial settings. Their high stability and resistance to degradation under harsh conditions further bolster their suitability for continuous catalytic processes, as evidenced by their successful application in the reduction of nitroarenes and the oxidative polymerization of aromatic compounds—reactions traditionally reliant on metal catalysts.

Yadav et al. [218] provided further evidence of RAOPs' versatility by developing an IPOP based on a twisted biphenyl scaffold, synthesized via the Menshutkin reaction

between 3,3',5,5'-tetrakis(bromomethyl)-2,2',6,6'-tetramethylbiphenyl (TMBB) and 4,4'-bipyridine (BPY). This IPOP, characterized by high thermal stability and insolubility, mimics the properties of phase-transfer catalysts like tetrabutylammonium bromide (TBAB). The authors demonstrated its efficacy as a recyclable heterogeneous catalyst in Michael addition reactions for synthesizing β -arylthioiketones and Knoevenagel condensation for biscoumarins, achieving excellent yields under mild conditions. The polymer's cationic sites and bromide anions facilitated these transformations, and its recyclability was confirmed over 10 cycles without loss of activity. This work underscores RAOPs' potential as metal-free, sustainable alternatives to conventional catalysts, contributing to the synthesis of biologically relevant compounds with a green approach.

Despite their numerous advantages, RAOPs face challenges that must be addressed to realize their full potential in organic synthesis [219–223]. Limited conductivity, poor solubility, and difficulties in characterizing active sites within the polymer framework can impede their widespread adoption. However, ongoing advancements in synthetic methodologies, characterization techniques, and computational modeling are steadily overcoming these limitations. RAOPs are poised to play a pivotal role in the evolution of sustainable catalytic systems. Their ability to mediate complex organic transformations, enhance energy storage, and facilitate stereoselective synthesis under environmentally benign conditions positions them as a cornerstone of future developments in both academic research and industrial applications [224]. As research progresses, RAOPs are likely to further bridge the gap between efficiency and sustainability, revolutionizing the synthesis of complex organic molecules and beyond. Building on the advancements in RAOPs, recent studies have further expanded the scope of sustainable catalytic systems, Ma et al. [225] reported the synthesis and application of fully conjugated COF nanosheets functionalized with carboxyl groups for visible-light-driven photocatalytic organic synthesis in water. The authors employed a cooperative strategy involving chemical exfoliation and group transformation to produce COF-JLU100COOHNSs, which exhibited enhanced dispersibility in water and prevented re-stacking due to the introduced carboxyl groups (Fig. 10A). These nanosheets demonstrated exceptional photocatalytic activity for the selective oxidation of sulfides and oxidative coupling of benzylamines under visible-light irradiation, outperforming state-of-the-art metal-free $g\text{-C}_3\text{N}_4$ nanosheets. The study highlighted the importance of the nanosheets' full π -conjugation, accessible active sites, and efficient charge carrier separation in achieving high photocatalytic efficiency. This work not only presents the first example of COF-based nanosheets catalyzing organic transformations in water but also expands the potential applications of ultrathin COF nanosheets in sustainable chemistry. The findings underscore the potential of functionalized COF nanosheets as versatile photocatalysts for green organic synthesis, aligning with the principles of environmental and energy sustainability. In a similar vein, Li et al. [226] developed a novel porous organic polymer-supported single-site nickel catalyst, POP-PP-Ni, for the efficient synthesis of poly-substituted pyrimidine derivatives via an acceptorless dehydrogenation strategy under solvent-free conditions. The catalyst was synthesized through a Friedel-Crafts reaction between 1,3,5-triphenyl benzene and 4,6-dichloro-2-(1 H-pyrazol-1-yl) pyrimidine, followed by metalation with nickel. The resulting material exhibited

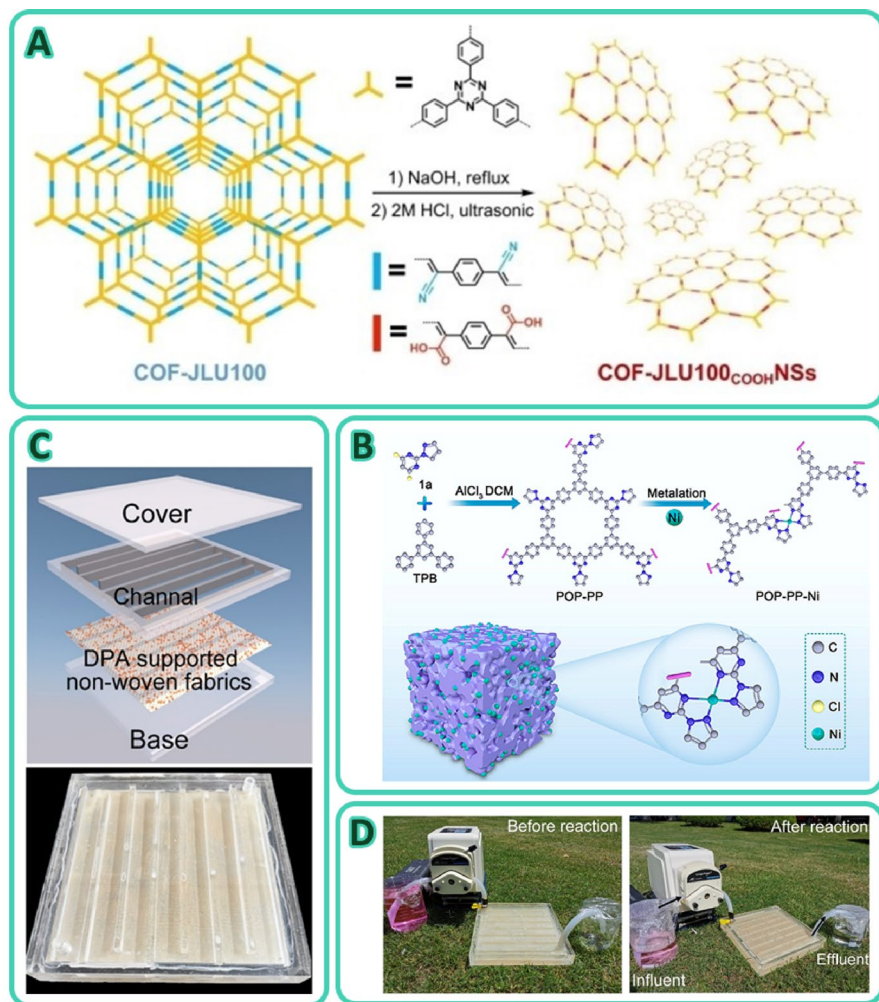


Fig. 10 (A) Synthesis process and structure scheme of COF-JLU100COOHNSs. Reprinted with permission [225]. (B) The synthesis of POP-PP-Ni. Reprinted with permission from [226]. (C) Schematic and physical drawings of the photocatalytic continuous-flow reactor. Reprinted with permission from [227]. (D) Photocatalysis-self-Fenton degradation of RhB over continuous-flow reactor before and after reactions. Reprinted with permission from [227]

a high surface area and hierarchical porosity, which facilitated excellent catalytic activity and stability (Fig. 10B). The authors demonstrated that the well-defined and isolated coordination sites of the polymer effectively immobilized nickel atoms, enabling high catalytic efficiency in the one-pot synthesis of pyrimidines from primary and secondary alcohols and benzamide oximes. Mechanistic studies revealed that the reaction proceeds through a sequence of acceptorless dehydrogenation, metal-mediated Michael reaction, and 1,5-electrocyclization. Notably, the catalyst maintained its activity over multiple cycles, highlighting its potential for sustainable and recyclable catalytic processes. This work provides a significant advancement in

the field of heterogeneous catalysis, offering a green and efficient method for synthesizing biologically relevant pyrimidine derivatives. Furthering the exploration of sustainable catalytic systems, Zhang et al. [227] developed a novel donor-polymeric carbon nitride-acceptor (DPA) catalyst for efficient sunlight-driven H_2O_2 synthesis and in-situ utilization. The researchers grafted spatially isolated donor (methoxyphenyl) and acceptor (anthraquinone) units onto the edges of polymeric carbon nitride, creating dual active centers that facilitate directional electron-hole transfer (Fig. 10C). This design enhances the two-electron oxygen reduction reaction (2e^- ORR) and four-electron water oxidation reaction (4e^- WOR), significantly boosting H_2O_2 production. The anthraquinone unit also exhibits strong proton extraction capabilities, further promoting H_2O_2 generation. The DPA catalyst demonstrated remarkable photocatalytic efficiency, surpassing traditional catalysts, and was successfully applied in a continuous-flow self-Fenton reactor for the degradation of organic pollutants under sunlight (Fig. 10D). This work presents a groundbreaking approach to designing photocatalysts for sustainable H_2O_2 synthesis and environmental remediation, highlighting the potential of molecular engineering in advancing photocatalytic technologies.

Complementing these developments, Naccache et al. [228] explored the development of greener heterogeneous catalysts for biodiesel synthesis by engineering the surface chemistry and morphology of polymeric carbon nitride (PCN) materials. The study focused on synthesizing polymeric carbon nitride dots (PCNDs) and nanosheets (PCNNs) through a solid-phase reaction between urea and sodium citrate, with the aim of optimizing their catalytic performance in the transesterification of canola oil to biodiesel (Fig. 11A). By varying the precursor ratios, the researchers tailored the materials' morphology and surface properties, achieving high biodiesel conversion rates under mild reaction conditions. The catalysts demonstrated excellent reusability over multiple cycles, maintaining high efficiency. The work also delved into the reaction kinetics, proposing a pseudo-first-order mechanism and highlighting the role of nitrogenated and carboxylate functional groups in the catalytic process. This study underscores the potential of metal-free, cost-effective PCN-based nanomaterials as sustainable alternatives to conventional homogeneous and metal-based heterogeneous catalysts, paving the way for more environmentally friendly biodiesel production processes. In parallel, Aflak et al. [229] developed a sustainable and efficient catalytic system for the synthesis of 1,2,3-triazole derivatives using a magnetic polyvinylpyrrolidone (PVPP)-supported copper(I) catalyst. The study aimed to address the growing demand for environmentally benign chemical processes by leveraging green chemistry principles (Fig. 11B). The researchers synthesized a $\text{Cu(I)}/\text{PVPP-Fe}_3\text{O}_4$ composite through a two-step process involving the in situ co-precipitation of Fe_3O_4 nanoparticles within a PVPP matrix, followed by the immobilization of copper(I) ions. The catalyst demonstrated high efficiency in copper-catalyzed azide-alkyne cycloaddition (CuAAC) reactions, enabling the regioselective formation of 1,4-disubstituted-1,2,3-triazoles in water at room temperature. The magnetic properties of the catalyst facilitated easy separation and reuse, enhancing the sustainability of the process. The study also evaluated the greenness of the catalytic system using metrics such as atom economy, E-factor, and EcoScale, highlighting its low waste generation and high sustainability scores. Overall, the

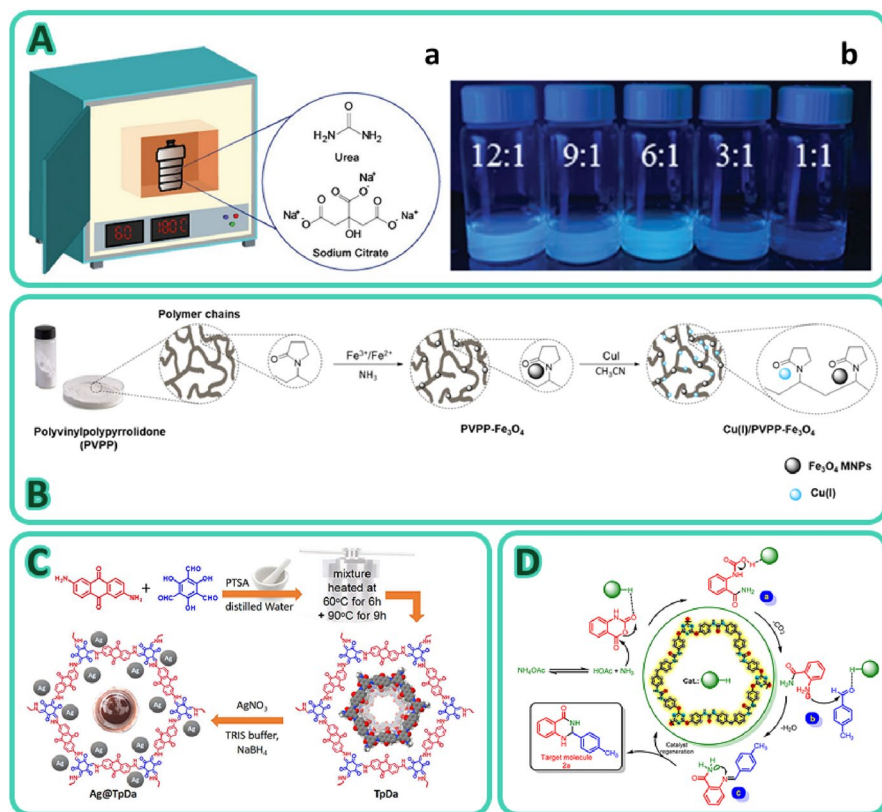


Fig. 11 (A) (a) TEM digital images for the catalysts displayed two different morphologies. (b) Showing blue fluorescence under UV excitation. Reprinted with permission from [228]. **(B)** Preparative route of Cu(I)/PVPP-Fe₃O₄ composites. Reproduced from Ref [229]. with permission from the Royal Society of Chemistry. **(C)** Schematic Presentation of the Preparation of Nanoporous Ag@TpDa. Reprinted with permission from [230] Copyright 2024 American Chemical Society. **(D)** A plausible mechanism for the preparation of 2f in the presence of urea-rich POP as catalyst. Reproduced from Ref [231]. with permission from the Royal Society of Chemistry

work presents a significant advancement in the development of reusable, eco-friendly catalysts for click chemistry, aligning with the principles of green chemistry and the United Nations Sustainable Development Goals. Expanding the scope of CO₂ utilization in organic synthesis, Islam et al. [230] explored the development of a sustainable and efficient catalytic system for the synthesis of tetramic and tetric acids using CO₂ as a C1 feedstock. The authors designed a silver nanoparticle-decorated covalent organic framework (Ag@TpDa) as a heterogeneous catalyst, which demonstrated remarkable efficiency in facilitating the carboxylative cyclization of unsaturated amines and alcohols under mild conditions. The study aimed to address the challenges associated with CO₂ activation and its utilization in organic synthesis, particularly in the production of biologically significant compounds such as tetramic and tetric acids, which are known for their diverse pharmacological activities (Fig. 11C). The Ag@TpDa catalyst not only exhibited high catalytic activity and selectiv-

ity but also showcased excellent recyclability, maintaining consistent performance over multiple cycles. Theoretical calculations using DFT were employed to elucidate the reaction mechanism, providing insights into the high selectivity and efficiency of the catalyst. This work highlights the potential of metal-supported covalent organic frameworks in sustainable chemical transformations, offering a green pathway for CO₂ fixation and the synthesis of valuable organic compounds. Furthermore, Zarei et al. [231] synthesized a novel urea-rich porous organic polymer (urea-rich POP) via a solvothermal method and investigated its potential as a hydrogen bond catalyst for organic transformations, specifically the Knoevenagel condensation reaction and the synthesis of 2,3-dihydroquinazolin-4(1 H)-ones. The study highlighted the polymer's ability to facilitate hydrogen bonding interactions, which significantly enhanced its catalytic efficiency in the targeted reactions (Fig. 11D). The urea-rich POP demonstrated excellent performance under solvent-free conditions, offering high yields and short reaction times for a variety of substrates. Additionally, the catalyst exhibited good reusability, maintaining its activity over multiple cycles. The work underscores the potential of urea-rich POPs as versatile and sustainable catalysts in organic synthesis, contributing to the development of efficient heterogeneous catalytic systems for environmentally friendly chemical processes. Moreover, Recent work by Su et al. [232] introduced a new class of ultrastiff microporous redox-active polymer aerogels (RAPAs), offering a significant leap in photocatalytic performance through a quasi-homogeneous reaction mechanism. These RAPAs, derived from poly(triazine imide) frameworks, exhibit extended π -conjugation, high redox tunability, and exceptional charge separation efficiency. Their porous structure facilitates rapid mass transport, while the mechanical stiffness ensures stability during photoredox processes. Notably, Su et al. demonstrated that these materials enable long-lived excited states and efficient electron transfer, paving the way for advanced solar-driven redox catalysis and hybrid energy conversion platforms.

RAOPs are increasingly gaining attention in the field of organic synthesis as efficient, tunable, and sustainable catalysts. Their modular design allows for precise incorporation of catalytic functionalities, such as redox-active heterocycles, metal-binding ligands, and electron-donating/withdrawing groups, enabling high activity and selectivity across a range of organic transformations. Their macromolecular nature also facilitates easy separation and reuse, promoting green and scalable synthetic protocols. Recent advances have shown RAOPs to be effective in C–C coupling, oxidation, hydrogenation, and other key reactions, often mimicking enzymatic or transition-metal-based systems. However, limitations remain in terms of catalytic turnover numbers, substrate scope, and mechanistic understanding. Future progress will depend on the development of structure–function relationships, improved synthetic routes for polymer backbones, and synergistic integration with other catalytic platforms. Overall, RAOPs offer a promising and sustainable route for advancing precision catalysis in modern organic synthesis.

Conclusion remarks and future perspectives

RAOPs have emerged as a revolutionary class of materials that exhibit promising applications in energy storage, energy harvesting, and catalysis. Their intrinsic properties, including tunable redox potential, lightweight nature, flexibility, and sustainability, position them as viable alternatives to conventional inorganic materials in next-generation energy technologies. Despite significant advancements in polymer chemistry, material design, and device integration, several challenges must be addressed to unlock the full potential of RAOPs and achieve large-scale commercialization. In energy storage applications, RAOPs have demonstrated impressive performance in lithium-ion, sodium-ion, and potassium-ion batteries, as well as supercapacitors. Their ability to undergo reversible redox reactions provides high charge-discharge efficiency and cycling stability. Despite these advantages, several challenges remain that must be addressed to fully realize the potential of RAOPs in commercial applications.

One of the key areas for future research is the enhancement of the electrochemical stability and conductivity of RAOPs. While recent studies have demonstrated promising performance improvements through molecular design strategies such as conjugated backbones, redox-active moieties, and hybrid composite materials, further optimization is necessary to enhance charge transfer efficiency and cycle life. Incorporating conductive additives, functionalized nanostructures, and metal-organic hybrid frameworks could provide novel approaches to overcoming the intrinsic limitations of RAOPs in energy storage applications. In addition to improving conductivity, addressing the scalability of RAOP synthesis is crucial for industrial adoption. Many RAOPs are synthesized via complex polymerization techniques that require precise reaction conditions and costly precursors. Developing cost-effective, large-scale production methods, such as green solvent processing, scalable electrochemical polymerization, and 3D-printing techniques, will be essential to bridge the gap between laboratory research and practical implementation. Another important consideration is the environmental sustainability of RAOPs. As global demand for renewable energy solutions increases, ensuring the biodegradability and recyclability of RAOP-based materials will be a significant factor in their widespread adoption. Designing RAOPs with bio-derived precursors, developing closed-loop recycling systems, and integrating these materials into eco-friendly energy storage solutions will further enhance their long-term viability. In the broader scope of energy storage and conversion, RAOPs could play a critical role in next-generation hybrid energy storage systems. The development of self-charging photo-supercapacitors, organic redox-flow batteries, and hybrid capacitive devices could leverage the unique properties of RAOPs to create highly efficient, durable, and sustainable energy solutions. The incorporation of these polymers into multifunctional devices, such as flexible and wearable electronics, could further expand their applications beyond conventional energy storage. Moreover, the synergy between RAOPs and catalysis presents an exciting avenue for future research. RAOP-based catalysts have shown remarkable efficiency in facilitating redox reactions, including water splitting, CO₂ reduction, and hydrogen evolution. By designing highly porous, conductive RAOPs with tailored redox-active sites, researchers can develop next-generation catalysts that

rival or surpass traditional metal-based systems in performance and sustainability. While significant strides have been made in advancing RAOP technologies, challenges such as long-term durability, charge carrier mobility, and industrial scalability remain key obstacles. Addressing these issues through continued research in materials chemistry, process engineering, and system integration will be crucial to unlocking the full potential of RAOPs.

Author contributions Farimah Kamran and Hana Afshar: Writing - Original Draft, Visualization, Investigation; Farangis Shahi: Conceptualization, Writing - Review & Editing, Supervision.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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